ScrollFiesta: A Virtual Meshing and Unwrapping Toolkit for the Herculaneum Papyri

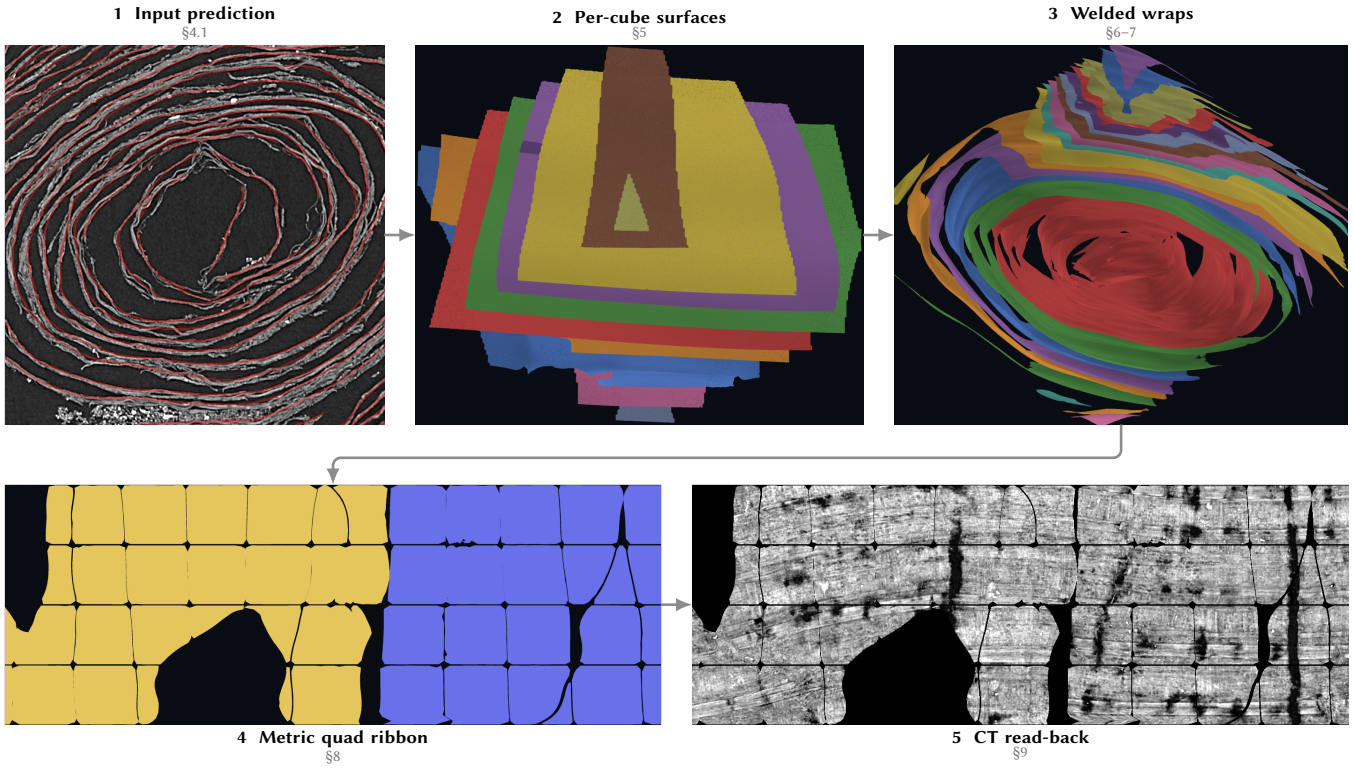


Fig. 1. **August research pipeline, on one PHerc0139 $4 \times 5 \times 5$ block.** The September public-configuration result appears separately in Figure 10. (1) The input: a completed coarse surface prediction (red) over raw CT at z=4416. Producing that volume without inventing a connection onto a neighbouring turn is the subject of the companion *Socratic Method* report; ScrollFiesta takes it as given (Section 4.1). (2) One 128^3 cube after moving-least-squares collapse, guarded ball pivoting, and topological cleanup, with each extracted sheet coloured separately. The sheets stay distinct components, so no two wraps in this cube are fused into one surface. (3) The welded block seen down the scroll axis. Welding joins fronts across cube seams only where both cubes agree on the wrap, so the wraps survive as long, separately identified spirals rather than one merged surface. (4) A window of the solved flat atlas, coloured by atlas group and cut into metric quad cells. (5) The same window of the CT read-back — the papyrus those exact cells carry. Black is absent material rather than missing texture: no layer supplies papyrus there, and the pipeline leaves the hole visible rather than interpolating across it. Figure 2 gives the same pipeline as a data-flow diagram.

Virtually unwrapping a carbonized Herculaneum papyrus poses a geometry problem of an unusual shape. The target is not the boundary of a solid but a single open sheet, many meters long, wound into a hundred or more turns that touch, fuse, delaminate, and tear. Standard machinery fits it badly: iso-surfacing and Poisson reconstruction recover closed, two-sided boundaries; a flattener minimizes distortion against whatever mesh it is handed, so two fused turns flatten into one clean page; and the freedom an atlas packer relies on — place a chart wherever space allows — is exactly the freedom this problem lacks, because horizontal position on the page *is* developed arc length, shared by every fragment of a turn and fixed only up to a whole number of turns.

We present *ScrollFiesta*, a meshing and unwrapping toolkit organized around one geometric instrument: a branch-cut-free winding coordinate that measures, for any two samples, whether they lie on the same turn of the roll or on different ones. It reappears as the decision variable at every scale, and it is the quantity no mesh-quality measure can see.

ScrollFiesta takes a completed surface-probability volume as its input; producing that volume is the subject of the companion *Socratic Method* report, and is deliberately out of scope here. We contribute three pieces. *First*, we separate local geometry from large-scale identity. Geometry is extracted per cube by an iterated moving-least-squares collapse, ball pivoting guarded by anti-parallel and accumulated-winding tests, topology repair, and CVT remeshing; identity is carried independently as sheet tracks whose pairwise integer winding relations compose through a maximum-support forest with integer subtree-shift loop closure, so geometry is never fused to make an order graph connected. *Second*, winding is inferred rather than obtained by rounding one fragile measurement. A two-sided generalized winding number field supplies noisy absolute per-component observations, local continuation and radial-order measurements supply pairwise factors, and an integer-label Markov random field reconciles both by alpha–beta swap graph cuts. *Third*, the accepted geometry is fitted to a ribbon of quadrilaterals whose horizontal coordinate is exact developed arc length, reconciled by Huber-IRLS sparse

least squares, untangled by exact triangle-contact response, and composited against CT.

In the August research campaigns, on a fresh $4 \times 5 \times 5$ block the winding MRF resolves 882,219 vertices in 10.8 seconds, lowering energy from 221.37 to 200.49 at continuation/order satisfaction 0.9997/0.9982; sheet books carry over 70% of evidenced material at every rung through $21 \times 21 \times 21$, ordering 15,457 tracks across 112 turns; and the metric fit conserves all 4.03M faces of the $4 \times 5 \times 5$ ribbon in 9.6 seconds. We also report the negative results that shaped the design, including a slab splitter that separates 0 of 246 fused runs and an overlap-minimizing tabu atlas that removes 99.5% of UV overlaps while tearing 41 weld relations — both instances of a geometric objective being asked to decide a question geometry cannot see.

The September public release removes tuning and automation frameworks and bottom-up hierarchical runners, preserves the versioned C API, and supplies one generic configuration. Its separate $4 \times 5 \times 5$ native sheet-assembly check preserves the represented source geometry and produces a CT-textured partial sheet at 47.4% source-area coverage, with 171 overlapping triangle pairs. We report that result as unqualified and retain the earlier research results with their original scope.

1 Introduction

A library of papyrus scrolls was carbonized at Herculaneum during the AD 79 eruption of Mount Vesuvius. The rolls survive, but they cannot be opened: the papyrus has become fragile carbon, and the writing is trapped inside a tightly wound coil. Synchrotron micro-CT records the internal geometry without touching the object, and the text becomes readable only after the correct papyrus surface has been located in three dimensions and *virtually unwrapped* to a plane.

From the perspective of geometry processing, the computational object virtual unwrapping requires is unusual. Scroll CT data does not represent the boundary of a solid; instead, it is a single open sheet, many meters long, rolled into a hundred or more turns that in places touch, fuse, delaminate, and tear. A useful reconstruction has to keep neighbouring turns distinct, distinguish a gap in the evidence from a real hole in the papyrus, and preserve physical length so that fibres and writing stay continuous after flattening. Getting one of those wrong does not produce a slightly worse page; it produces a confidently wrong one.

Automatic U-Net segmenters [Ronneberger et al. 2015] supply strong voxel evidence but not that object. Their output can be several voxels thick, discontinuous across dropouts, or fused across adjacent wraps where the sheets are pressed together. Classical surface reconstruction is also not suited to this data: marching cubes [Lorensen and Cline 1998] and Poisson reconstruction [Kazhdan et al. 2006] are built to recover a closed, two-sided boundary, while a scroll needs a single-sided open manifold. Nor can parameterization rescue the result afterwards. A flattener minimizes distortion against whatever mesh it is given [Sheffer et al. 2005], so two wraps fused into one surface flatten into one clean page; and the freedom an atlas packer relies on — place charts wherever empty space allows [Limper et al. 2018] — does not exist here, because horizontal position on the page *is* the answer: developed arc length, shared by every disconnected fragment of a wrap, and fixed only up to a whole number of turns that no continuous relaxation can recover. Worse, the failures at all three stages are often invisible when rendered: a short connection that happens to walk a full turn, a chart placed on the wrong wrap, or a fabricated continuation through a dark region of the CT all look like clean geometry. They are only detectable by asking what evidence supports them.

ScrollFiesta is a meshing and unwrapping toolchain built for the Herculaneum papyri. It takes a coarse surface-probability volume, the matching CT, and an estimate of the scroll axis, and works in two domains (Fig. 2).

The organizing idea is a single geometric quantity. Because the object is a spiral, the branch-cut-free winding coordinate of Equation (1) — how many turns' worth of radius a point has climbed, minus how far around the roll it has travelled — is constant along one wrap and increases by exactly one on stepping outward to the next. It is therefore the natural coordinate in which "same sheet" and "next sheet" differ by a unit, and it is precisely the quantity that edge length, aspect ratio, manifoldness, and self-intersection cannot measure. Every stage below carries it: it guards a pivot in ball pivoting, defines the relation graph between sheet tracks, becomes the label set of a Markov random field, and finally becomes the horizontal axis of the flattened page.

Its input is a completed surface-probability volume. Repairing that volume — distilling fine-resolution evidence into the coarse predictor, and closing short dropouts in the traced sheet — is the subject of the companion *Socratic Method* report [ScrollFiesta 2026], and everything below assumes it has already been done. ScrollFiesta itself adds and removes no probability mass; it decides what surface the volume it is handed describes.

First, in the identity domain (Section 7), the prediction is traced into material tracks and lifted into single-valued sheets. Signed radial contacts between coexisting tracks give integer winding relations, which compose through a maximum-support forest repaired by integer subtree shifts — an identity structure that is deliberately not mesh connectivity, since two pieces of one sheet across a hole are disconnected and two different wraps joined by one fusion are connected.

Second, in the metric domain (Sections 5–9), local geometry is extracted cube by cube as single-sided open meshes; a two-sided generalized winding number field and local component relations are reconciled as uncertain factors in an integer Markov random field; and the resulting geometry is fitted to a ribbon of quadrilaterals whose horizontal coordinate is exact developed arc length.

Contributions. Relative to the July 31 public release, this update contributes:

(1) an identity structure that is deliberately *not* mesh connectivity: sheet tracks carrying pairwise integer winding relations, composed through a maximum-support forest with integer subtree-shift loop closure, so that two fragments of one sheet separated by a hole remain one identity while two wraps joined by a single fusion do not become one;
(2) an inference formulation for winding that replaces hard rounding of one fragile measurement: a two-sided generalized winding number field yields robust per-component unaries, local continuation and radial-order observations become pairwise factors, and alpha–beta swap graph cuts solve the resulting integer-label MRF over residual turn offsets; and

(3) a metric quad ribbon whose horizontal coordinate is exact developed arc length rather than the output of a distortion-minimizing parameterization — fitted against one shared metric-rate map, reconciled by Huber-IRLS sparse least squares, untangled by exact triangle-contact response, and composited against held-out CT.

Scope. This is a progress report, not a claim that a Herculaneum scroll has been read end to end. The winding MRF improves its objective on a fresh $4 \times 5 \times 5$ mesh, but the sparse conflict penalties layered on top of it do not yet reduce duplicate claims without damaging the relation evidence they are built from. The sheet-track ladder runs through $21 \times 21 \times 21$ cubes; the metric ribbon is measured at $4 \times 5 \times 5$, and the larger metric numbers come from separately frozen upstream states rather than one controlled scaling curve. Each such limit is stated with the result it belongs to.

How to read this paper. Section 2 is a two-way glossary, covering both geometric mesh processing and the scrolls themselves, and should be read first by anybody who is not familiar with one half of the problem. A reader arriving from geometry processing who has never handled a scroll wants Sections 2.1 and 2.2; a reader who knows the scrolls but not the reconstruction literature wants Section 2.3, which states exactly which standard assumptions this object violates. Section 4 gives the data flow stage by stage. Sections 5 to 9 then take those stages in order, each opening with a plain statement of what the stage does physically before the technical account begins. Section 12 collects the measurements, including the negative results. Appendix A is the only part of the paper organized by date: it records what changed since the July 31, 2026 public release.

2 Background: The Object, the Data, and the Coordinates

This section fixes the vocabulary used throughout the paper, in both directions. Sections 2.1 and 2.2 describe the physical object and its digital representation for a reader who arrives from geometry processing and has never handled a scroll. Section 2.3 restates the same problem in the language of surface reconstruction and parameterization for a reader who knows the scrolls but not that literature, and says precisely which standard assumptions this object violates. Section 2.4 defines the three coordinate systems both audiences need, and Section 2.5 names the four failures the rest of the paper is built to prevent. A reader fluent in both fields can go to Section 3 and use Table 1 as a reference.

2.1 The Physical Object

A Herculaneum book is one continuous papyrus sheet, tens of centimetres tall and several metres long, rolled around a central void. Carbonization has left it as brittle carbon at nearly the density of the surrounding matrix, so in CT the sheet appears as a faint, thin, low-contrast structure rather than an obviously separable material.

We use the following physical terms consistently.

Wrap (turn). One circuit of the sheet around the roll. The whole scroll is one sheet visiting many wraps in succession. Two points can be a fraction of a millimetre apart in space and metres apart along the sheet if they sit on adjacent wraps. This is the central difficulty of the problem.

Pitch. The radial distance from one wrap to the next, measured outward from the centre. On PHerc0139 the calibrated pitch is approximately 9.5 coarse voxels and the physical clearance between neighbouring wraps is approximately 7 voxels. Pitch is not constant: it varies with radius, with local crushing, and with delamination, which is why the system measures it locally rather than assuming one global value.

Umbilicus. The central axis of the roll, and the origin from which radius is measured. Near it the wraps are crushed together, the pitch is not measurable, and the prediction is unreliable; ScrollFiesta excludes material inside radius 64 for this reason.

Recto and verso. The two faces of the sheet. Only one carries the writing of interest, so the reconstruction must be *single-sided* and consistently oriented. A surface that flips from recto to verso partway along is unusable even when it is geometrically perfect.

Delamination. The sheet has split through its thickness into two nearly coincident plies. Both are real papyrus and both may carry evidence, so they must be separated and kept, not averaged into one blurred layer.

Fusion. Two *different* wraps have been pressed into contact, or the segmenter has connected them. This is the catastrophic failure: a fused pair reads as a single sheet, and every downstream stage inherits the error.

Dropout. The papyrus is present but the segmenter failed to report it, typically across a few voxels of a faint region. A dropout should be repaired. A genuine tear should not.

2.2 The Data

CT volumes. Two resolutions are used. The *coarse* production volumes are at 7.91 or 9.362 μm per voxel; these are the grids on which meshing, unwrapping, and texture read-back are performed, and one coarse voxel is the unit of length throughout this paper unless stated otherwise. A *fine* volume at 2.399 μm per voxel exists for parts of some scrolls. Fine CT holds roughly sixty times more samples per unit volume, so it is never a production substrate, and the geometry described in this paper never reads it. It is used as *privileged evidence* by the companion *Socratic Method* pipeline, which judges its prediction-volume decisions against it before that volume reaches us.

Prediction volume. A 3-D U-Net — the publicly released "M7" recto-surface model — assigns every coarse voxel a probability of lying on the recto surface. Thresholding that field gives the *prediction*, a binary volume which is our primary input. It is not a surface. It is several voxels thick where the model is confident, absent where it is not, and occasionally welded across wraps.

RAW.. The unmodified CT intensity volume, as distinct from the prediction. RAW is the authority on what the papyrus actually looks like and is the source sampled to produce the final readable image. Throughout the pipeline it is held back from stages that would

otherwise fit to it, and consulted afterwards as an independent check.

Cube grid. The volumes are far larger than memory, so the region of interest is partitioned into a regular grid of cubes, each processed independently with a small overlapping *halo* of shared voxels so that neighbouring cubes agree along their common boundary. Grid sizes in this paper are written $4 \times 5 \times 5$, $10 \times 10 \times 10$, $4 \times 21 \times 21$, and $21 \times 21 \times 21$. These form the ladder of scales on which every result is reported, and a rung must pass before the next is attempted.

2.3 The Same Problem in Geometry-Processing Terms

Stated as a reconstruction and parameterization problem, the scroll violates six assumptions that most of the standard toolchain is built on. Naming them is the fastest way to see why so much of this paper is spent on an explicit winding coordinate rather than on energies.

(1) **The target is an open, single-sided manifold, not a closed boundary.** We are recovering one side of a sheet with a genuine free boundary (its torn edges), embedded in $\mathbb{R}^3$. Isosurface and indicator-function methods [Kazhdan et al. 2006; Lorensen and Cline 1998] return a watertight two-sided surface, which here means recto welded to verso around every rim. There is also no "inside" to define, so an implicit formulation has no sign to fit.

(2) **Proximity is not adjacency, and the gap is smaller than the noise.** The surface passes within roughly 7 voxels of *itself*, once per turn, over its entire extent. Every neighbourhood query, ball radius, normal estimate, and Hausdorff-style tolerance is therefore adversarial: the nearest point in space is very often the wrong point on the sheet. Radii cannot be tuned generously, because the safe band is bounded above by the inter-wrap clearance and below by the sample spacing, and those two bounds are within a factor of a few of each other.

(3) **The decisive error is invisible to every geometric quality measure.** A triangle strip bridging two adjacent wraps has good aspect ratio, consistent orientation, small dihedral angles, low curvature, and no self-intersection. Edge length, degeneracy, manifoldness, normal-flip, and Euler-characteristic checks all pass it. In one measured case, 130,499 interpolation cells were generated across such a bridge, 24.4% of them spanning more than a full inter-sheet pitch. The correctness of a connection is simply not a function of local geometry, which is why this system carries a separate winding coordinate (Section 2.4) rather than trusting a mesh-quality score.

(4) **The parameterization carries an integer degree of freedom per component.** Local flattening determines each connected chart up to a whole number of turns. That integer is not recoverable by any continuous relaxation — the energy landscape is flat between admissible values and steeply wrong between them — so winding must be assigned by discrete inference over measured relations. Historically we attacked it with local search; the current system assigns it from radial evidence and a maximum-support forest instead (Section 7), for reasons Section 12.5 makes measurable.

(5) **The map wanted is isometric, not conformal, and its domain is extreme.** The horizontal coordinate must be true developed arc length, because a reader is comparing letter shapes and fibre continuity across metres of sheet; angle-preserving distortion of a few percent accumulates into unreadable drift. The output domain is a strip of extreme aspect ratio — $31{,}127 \times 508$ pixels for a $4 \times 5 \times 5$ region, $33{,}338 \times 1{,}276$ for $10 \times 10 \times 10$ — which no conventional atlas packer is designed to produce, and which is not free to be repacked, because horizontal position *is* the answer.

(6) **The data do not fit in memory, and correctness must survive being cut up.** A production volume is on the order of 10^{11} voxels. Everything is computed per cube and composed, so every operator must be local enough that two cubes sharing a halo produce matching geometry along their seam — a constraint that rules out several otherwise attractive projections and simplifiers, and that motivates the compact-support choices in Section 5.

Two consequences run through the whole design. First, because no geometric measure can see the decisive error, the system carries non-geometric information — winding phase above all — alongside the mesh, and decides on that. Second, because the safe operating band is narrow and bounded by physical constants of the object, most parameters here are not tuned for appearance; they are derived from measured pitch and clearance, and Appendix C records them with the measurement that fixed each one.

2.4 Three Coordinate Systems

Nearly every decision in this paper is a statement in one of three coordinate systems, and most confusion about scroll reconstruction comes from moving a quantity between them without saying so.

World coordinates (x, y, z). Voxel positions in the scan. The axis of the roll lies close to z, so z runs along the height of the scroll and a slice at constant z cuts the roll into a spiral cross-section. World distance is the wrong measure of separation on the sheet: two points one voxel apart may belong to different wraps.

Cylindrical coordinates (r, θ, z). Radius from the umbilicus, angle about it, and height. These express *where in the coil* a point sits. The important derived quantity is the winding number: for a local pitch p and winding sense $s \in \{-1, +1\}$,

$$w \;=\; s\,\frac{r}{p} \;-\; \frac{\theta}{2\pi}, \tag{1}$$

lifted continuously rather than wrapped into a principal range. Read plainly, the first term counts how many turns' worth of radius a point has climbed and the second subtracts how far around the roll it has travelled. On an ideal spiral the two cancel, so w is constant along one wrap and increases by exactly 1 each time one steps outward to the next. Equation (1) is therefore the basic instrument for distinguishing a same-wrap relationship from a next-wrap one, and it reappears as a guard in the mesher (Section 5), the welder (Section 6), and the sheet assembler (Section 7). Between consecutive

samples i, j the branch-cut-free increment is

$$\delta w_{ij} \;=\; s\,\frac{r_j - r_i}{p} \;-\; \frac{\mathrm{unwrap}(\theta_j - \theta_i)}{2\pi},$$

which is zero along a correct spiral ramp and accumulates ±1 for every wrap a fusion crosses. Accumulating δw along a path, rather than testing each step in isolation, is what catches the failure that actually occurs in practice: a surface front that climbs a through-thickness wall gradually, every individual step innocent, gaining whole turns over a long distance.

Developed coordinates (u, v). The flattened page. Here u is *developed arc length* — the distance one would travel along the unrolled sheet, so that a full turn costs the local circumference — and v is position along the scroll axis. Producing a correct u is what virtual unwrapping means. Two requirements make this harder than ordinary surface parameterization: different wraps must never occupy the same (u, v), and disconnected observations of the same wrap must share one metric frame. Conventional atlas packing [Limper et al. 2018], which is free to place charts anywhere empty space allows, can express neither.

2.5 What Failure Looks Like

Three named failures recur throughout the paper. Each needs a different instrument, which is why the system measures several independent quantities rather than one quality score.

A *full-turn fusion* is a connection whose accumulated δw reaches ±1 or more: two different wraps have been joined into one surface. It is invisible to every geometric quality test, because a bridge paved between two wraps is smooth, non-degenerate, and orientation-consistent. Only winding phase, or an explicit record of which triangles were generated rather than measured, can see it. We detect it with the accumulated-phase tests of Sections 5 and 6 and with an independent full-turn mesh scan.

A *phantom* is fitted geometry that lands where there is no papyrus at all, typically a prediction artifact lifted into a plausible-looking sheet. It is detected by sampling held-out RAW *after* the geometry has been frozen: a phantom sits over black CT and fails to correlate better than a null placement.

Fabrication is generated material presented as observation — a hole closed by a smooth interpolant, a raster void painted over, a continuation drawn across a region where the source itself had fused two wraps. Fabrication cannot be detected after the fact, so it is instead kept visible: direct, interpolated, filled, and moved-by-CT labels ride on every vertex and pixel, so a coverage statistic cannot be improved by inventing material.

3 Related Work

Virtual unwrapping of the Herculaneum scrolls. Virtual unwrapping was introduced by Seales et al. [2016], who recovered text from the En-Gedi scroll by segmenting its surface within a CT volume and flattening it to a plane. Applying the same idea to the far more tightly wound Herculaneum scrolls has driven several automatic approaches collected by the Vesuvius Challenge. *Spiral fitting* [Henderson 2026] uses the fact that the scroll was originally one rolled sheet and fits an extruded spiral to the prediction by solving a variational problem; it degrades where a noisy or locally collapsed prediction cannot support a single spiral. *Surface tracing* grows local surface patches and must subsequently decide which fronts or patches belong together. *ThaumatoAnakalyptor* extracts a point cloud through image filtering, classifies points into wraps, and reconstructs a surface through winding-angle assignment.

These families expose the same underlying trade. A strong global scroll prior can cross damaged regions but imposes an idealization the object does not obey; purely local evidence represents folds and tears faithfully but leaves a difficult composition problem. ScrollFiesta uses both, and its contribution is largely about *where* each decision is made rather than about a new primitive. Local geometry is reconstructed per cube. But large-scale identity is represented as sheet tracks carrying an explicit winding relation — never inferred from whether two triangle soups happen to be weldable — and the developed ribbon then consumes those identities and the measured phase field. This separation is what lets the system leave a global continuation open while still placing the local material on either side of it.

Generalized winding fields and discrete labeling. Generalized winding numbers extend the solid-angle winding function to imperfect oriented triangle meshes [Jacobson et al. 2013]; hierarchical approximations make them practical on triangle soups [Barill et al. 2018], while a recent boundary-edge reduction provides an exact alternative [Xie et al. 2026]. Those works use the field primarily for inside–outside reasoning. We instead sample both sides of an open papyrus sheet: the mean of the two values is a noisy global

Table 1. Terms used throughout the paper.

Term	Meaning
Prediction	Thresholded U-Net surface-probability volume; the primary input, and not yet a surface.
RAW	Unmodified CT intensity volume; authority for the final image, held out from construction.
Voxel	One coarse CT sample (7.91 or 9.362 μm); the unit of length in this paper.
Cube / halo	Unit of divide-and-conquer processing / the shared border that makes neighbouring cubes agree.
Winding number w	Eq. (1): constant along one wrap, +1 per wrap outward.
Track	One piece of material followed across slices; the unit of identity.
Sheet	A track lifted into a single-valued 3-D surface and checked against held-out RAW.
Sheet book	The collection of accepted sheets, kept separate rather than fused into one mesh.
Chart / component	One connected piece of mesh.
Quad ribbon	The regular quadrilateral lattice fitted to accepted material; the flattening substrate.
Developed length u	Arc length along the unrolled sheet; the horizontal axis of the final page.
Sander distortion	Per-triangle measure of stretch introduced by flattening; 1.0 is isometric.
Quarantine	Suspension of a track's *order* relation while its geometry is retained.

turn cue, while their difference should be one and acts as a local reliability measurement. We do not round this cue into a label. It becomes a robust unary in an integer Markov random field whose pairwise factors retain every independently measured continuation and radial-order relation. Alpha–beta swap graph cuts [Boykov et al. 2001] optimize small residual label windows hierarchically; sparse exclusion penalties represent evidence that one particular component–turn assignment is impossible without pretending that a new Gaussian observation was made.

Surface reconstruction and meshing. Surface reconstruction is a mature field, but its standard tools are a poor fit here because they are designed to produce *two-sided, watertight* surfaces. Marching cubes [Lorensen and Cline 1998] and its descendants, such as dual contouring [Ju et al. 2002] and differentiable FlexiCubes [Shen et al. 2023], extract a closed isosurface. Poisson reconstruction [Kazhdan et al. 2006] fits a closed indicator function and, in distributed form [Kazhdan and Hoppe 2023], scales to very large volumes. A scroll sheet is instead open and single-sided, so ScrollFiesta combines a moving-least-squares collapse with guarded ball pivoting [Bernardini et al. 1999] and explicit topology repair (Section 5). The choice is not incidental: a closed-surface method has no vocabulary in which to leave two nearby sheets unconnected, which is precisely the configuration this data is full of.

Mesh parameterization and atlas construction. Flattening a surface is likewise mature, but classical objectives solve a different problem. Angle-based flattening and LSCM [Sheffer et al. 2005] minimize angular or conformal distortion without providing the global layout a long, damaged scroll requires. For a clean disk-like patch ScrollFiesta delegates to a symmetric-Dirichlet flattener [Smith and Schaefer 2015], chosen because it is locally injective by construction: on a long, high-aspect, damaged patch the failure that matters is a folded chart, not a few degrees of angular error. For a whole scroll, per-chart injectivity is only the beginning, for the two reasons given in Section 2.4.

StrokeStrip [Pagurek van Mossel et al. 2021] is the closest local parameterization model, in that it fits one common coordinate system to a cluster of strokes rather than to a single connected patch; ScrollFiesta adopts its slice-domain view. An earlier version of this system paired that view with a tabu search [Glover 1989] over integer chart depths, which removed 99.5% of exact UV overlaps but left 41 usable weld relations torn — a result we retain as a reproducible baseline (Appendix C.8) and discuss in Section 12.5. The current method keeps the slice-domain metric observations but replaces search over chart depths with inference over turn labels: a maximum-support forest initializes the frame, loop closure and the MRF reconsider every retained relation, and unresolved sites carry a confidence rather than a fabricated certainty. Robust sparse least squares then closes metric residuals; it does not choose which wrap a component belongs to. Exact 3-D collision response, CT attachment, and layer compositing are separate operators rather than additional terms in one atlas objective, precisely so that a low-overlap score cannot buy a wrong physical identity.

4 System Overview and Data Flow

Scope of this account. The two research lanes below describe the August 2026 system and its recorded experiments. The September public release adds a native chart assembler, removes the tuning and hierarchical runner frameworks, and ships one shared configuration. Section 10 documents that interface and its separate measured $4 \times 5 \times 5$ result.

4.1 Inputs and Outputs

ScrollFiesta consumes a *completed* coarse surface-probability volume and the matching CT, their world-space transforms, a cube grid, and an estimate of the scroll axis and umbilicus. The prediction volume is taken as given. It is the output of the companion *Socratic Method* pipeline [ScrollFiesta 2026], which distills fine-resolution evidence into the coarse predictor and closes short dropouts in the traced sheet before any geometry exists; nothing described below writes back into it.

The outputs are plural because the two domains answer different questions and a run may usefully stop at either of them: sheet tracks with their support and interpolation masks; a winding-order graph with per-component winding field, label, and confidence; per-cube or aggregated geometry; and a developed quad ribbon with its CT texture.

4.2 Five Requirements

Five properties of the problem shape everything that follows, and each is answered by a specific mechanism later in the paper.

(1) **The target is one side of an open sheet**, not a watertight boundary — hence guarded single-sided reconstruction rather than isosurfacing (Section 5).
(2) **The data exceed memory** and must compose from axial shards and per-cube geometry — hence compact-support operators whose results agree across a halo (Sections 5 and 6).
(3) **One full-turn shortcut can invalidate a global unwrapping**, and connectedness and manifoldness cannot detect it — hence an explicit winding coordinate, and generalized-winding observations reconciled with local relations in an integer MRF (Sections 7.2 and 8.1).
(4) **A neighbouring wrap may be bright and physically touching**, so intensity does not determine topology — hence a pivoting front guarded by accumulated winding phase rather than by proximity, and a weld that joins only where both sides agree on the wrap (Sections 5 and 6).
(5) **Interpolated geometry and measured geometry behave differently downstream**, so the two are labelled apart rather than merged into one coverage number (Sections 9 and 11).

4.3 The Data Flow

Figure 2 shows the flow. Read it as two lanes that share one input artifact.

The shared volume. The completed prediction is the hand-off from the companion pipeline, and both lanes read it independently.

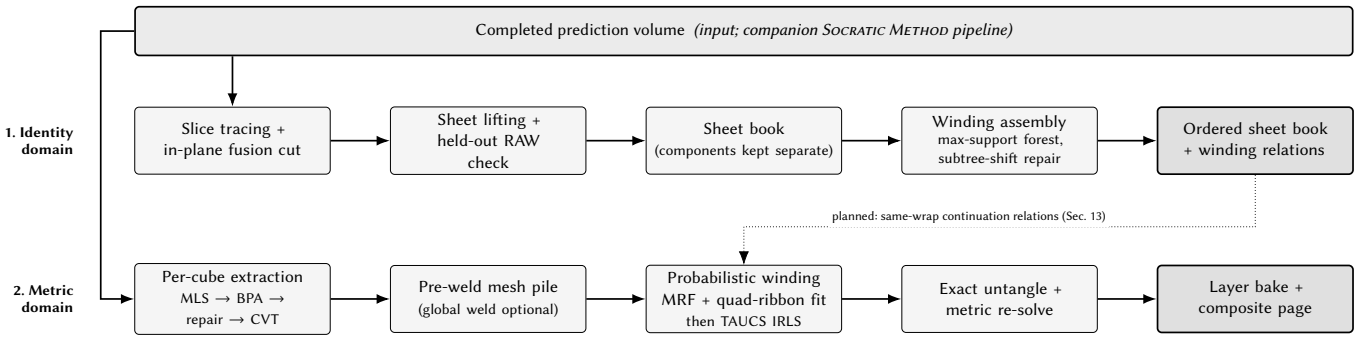


Fig. 2. The ScrollFiesta data flow. Solid arrows carry geometry and volumes; the dotted arrow is a link that does not yet exist. The completed prediction volume is an input rather than a stage: it is produced by the companion *Socratic Method* pipeline and is never written back to from here. The identity domain decides what is one sheet without fusing geometry. In the metric lane, generalized winding and local relations become uncertain factors in an integer MRF before metric fitting. Both lanes read the same volume independently, which is what makes a change to that volume measurable through either one in isolation.

Keeping the two lanes decoupled is what makes a prediction change measurable through either one in isolation.

Lane 1: the identity domain (Section 7). Here the system decides *what is one sheet* and *in what order the sheets are read*. The prediction is traced as a stack of two-dimensional paths; each path is associated across planes into a material track; a path that travels from one wrap through a fusion onto another is cut along its length and re-associated, so that no downstream index inherits a fused identity. Each candidate track is lifted into a single-valued sheet over its supported domain, and CT is consulted afterwards — the fit never sees it — to reject phantom sheets lying where the CT is empty. Accepted sheets form a *book*. Signed radial contacts between coexisting tracks produce integer winding relations, which are composed through a maximum-support forest and repaired by integer subtree shifts. The structural point is that a winding edge states an order relation and nothing else: geometry is never fused to make the order graph connected.

Lane 2: the metric domain (Sections 5–9). Here the system decides *where every point lands on the page*. Local geometry is extracted per cube by a moving-least-squares collapse, guarded ball pivoting, topology repair, and CVT remeshing. Before fitting, the oriented mesh soup is sampled on both sides to obtain generalized-winding means and jumps. Robust per-component unaries, same-sheet continuation factors, and radial-order factors are reconciled in a small-label MRF, which can be run on its own to obtain turn labels without any metric solve. The ribbon fit then intersects the soup with planes normal to the scroll axis and solves for a metric coordinate field; robust sparse least squares reconciles cracks and contacts; exact contact response untangles the coil; and layer-aware CT compositing emits the page.

What is not yet connected. The dotted arrow in Fig. 2 is deliberate. The sheet book currently supplies radial order for the traced material, and the metric lane runs its own winding registration on the geometry it receives. Feeding the book's same-wrap continuation evidence into the ribbon fit — without merging components — is the main open item of Section 13, and it is drawn as absent rather than implied.

4.4 Stage Summary

Table 2 gives each stage as what it consumes, what it produces, and the mechanism that does the work.

Two structural choices in this table are worth isolating, because they are what the winding coordinate buys. First, local geometry (rows 7–8) and large-scale identity (rows 5–6) are computed by different operators on different data, so a remeshing or welding decision cannot quietly change which turn a piece of papyrus belongs to. Second, the metric fit (row 9) *consumes* a turn assignment rather than searching for one, so a low-distortion or low-overlap solution cannot buy itself a different physical identity — which is exactly how the earlier overlap-minimizing atlas of Section 12.5 failed.

4.5 Current Experimental Boundary

The two lanes have not been carried to the same scale, and this paper does not present them as if they had. The sheet-book ladder runs through $21 \times 21 \times 21$. The winding MRF is measured on a fresh $4 \times 5 \times 5$ mesh. Fixed-topology metric projection has run on $4 \times 5 \times 5$, $10 \times 10 \times 10$, and $4 \times 21 \times 21$, but those larger inputs come from separately frozen upstream states and do not constitute one controlled end-to-end scaling curve. At $21 \times 21 \times 21$ the per-cube fleet and stage-1 concatenation are complete — 9,110 unique world-frame meshes contributing 195.971 million vertices and 381.764 million faces — but the global ribbon fit has not been run. Sections 12, A, and B keep these distinctions attached to every number.

5 Per-Cube Surface Extraction

In brief: the completed prediction is a slab of lit voxels, several voxels thick, not a surface. This stage turns each cube of it into a thin, consistently oriented, single-sided triangle mesh — and, more importantly, declines to connect anything it cannot justify. Every operation here is local, so that neighbouring cubes independently produce matching geometry along their shared border.

This is where the volume becomes geometry. Its outputs serve three consumers: verified local patches, cube-level inspection, and — for the active whole-sheet path — the pre-weld mesh pile consumed directly by the ribbon fit of Section 8. The operations below generate

Table 2. Stage summary. The right-hand column names the operator that does the work; the section reference gives the full account, and Appendix C records the constants.

Stage	Consumes	Produces	Principal mechanism
Sheet lifting (7.1)	completed prediction, material tracks	single-valued sheets, support masks	along-curve ownership test cuts a path that crosses a fusion onto another wrap; dense single-valued fit over the supported domain, with CT sampled only after the fit
Winding assembly (7.2)	sheets, local radial contacts	integer order graph	signed radial separation normalized by *local* pitch and snapped within 0.25 turns; maximum-support forest with integer subtree-shift loop closure
Probabilistic winding (8.1)	oriented mesh, GWN samples, local relations	integer turn labels, confidence	two-sided generalized winding number gives robust component unaries; continuation and order observations give pairwise factors; alpha–beta swap graph cuts over residual labels in $[-2, 2]$
Per-cube extraction (5)	prediction cube + halo	single-sided mesh charts	iterated MLS collapse onto the sheet; ball pivoting guarded by an anti-parallel test and accumulated winding phase; topology repair; CVT remeshing
Weld (6), optional	placed cube meshes	seam-joined mesh	seam-band refinement to ~3.5-voxel triangles, BPA bridging within a 0.35-turn phase tolerance, then mutual-best sheet correspondence across each seam
Quad-ribbon fit (8)	pre-weld mesh pile, measured phase	metric (u, v) field, material components	axis-normal slicing into chains; exact chain arclength against one shared reciprocal-integrated metric-rate map; Huber-IRLS sparse least squares over metric, contact, and crack rows
Untangle (9.2)	fitted ribbon	intersection-free geometry	exact triangle-contact set carrying phase branch and radial order; screened radial displacement field on an active collar under unilateral inter-turn clearance
CT attachment (9.3)	ribbon, RAW	moved vertices	winding-aware corridor search for the papyrus band, robust vector quilting, smooth displacement, and per-component accept/reject on measured collision, coverage, darkness, and distortion
Composite (9.4)	layer bakes	page image	hard layer conflicts split into separate bakes; first-cover ranking prefers direct coverage from any layer over an earlier layer's generated fill

local surface evidence. They do not establish large-scale wrap identity, and in particular the fact that two fronts can be connected is not treated as evidence that they should be.

5.1 Input Guard

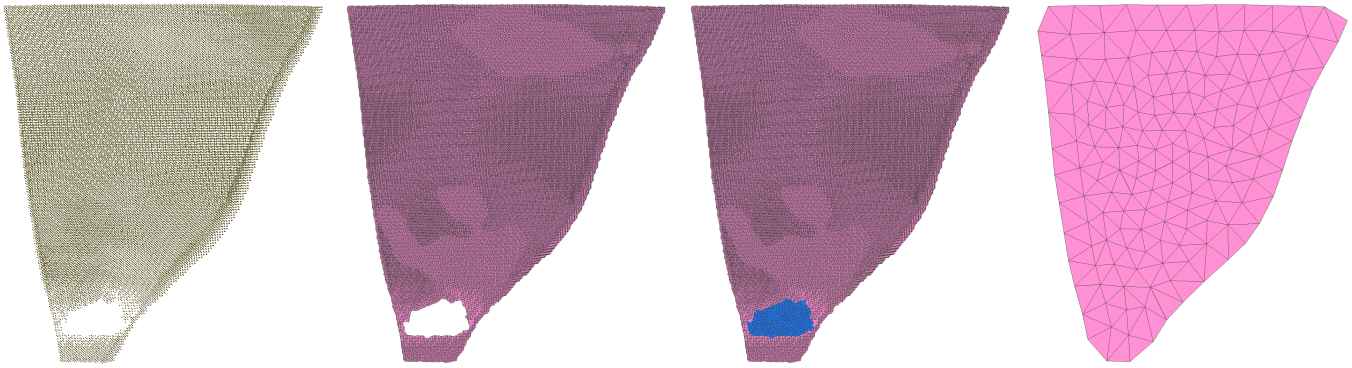
Before meshing a cube we apply a lightweight input guard. A U-Net occasionally emits a degenerate prediction in which a region is filled as a solid block rather than a thin sheet. Ball pivoting cannot roll across a solid three-dimensional point fill, and meshing such a cube yields only fragmented debris that pollutes everything downstream. We detect these cubes by morphological erosion-survival: a small 3-D erosion dissolves a genuine thin sheet entirely, while a solid slab retains a large interior fraction. We additionally require that the surviving foreground form a solid filled rectangle across many consecutive slices, so that dense but legitimate wrap regions — which are thin sheets pressed close together, not slabs — are never rejected. A cube failing this guard is skipped and contributes nothing, rather than contributing junk.

5.2 Collapse to a Single Sheet

We initialize the point cloud by taking one point at the centre of each solid voxel of the input cube. Where the prediction is more than one voxel thick this yields a slab of points several voxels deep, so we first collapse and denoise it into a single clean sheet using an iterated moving least-squares (MLS) projection [Levin 2003]: each point is moved along its local surface normal, estimated from a Wendland-weighted neighbourhood [Wendland 1995], onto the local least-squares plane. Only the through-thickness component of this motion is non-zero, so the slab is thinned without the in-plane contraction we observed when we initially tried the closely related locally optimal projection [Lipman et al. 2007].

The projection itself is a standard tool. Our observation is that choosing a projection whose kernel has *compact support* makes the per-cube result consistent across cube boundaries, and that this boundary-consistency is precisely what makes divide-and-conquer scroll meshing possible at all. Concretely, adjacent cubes are processed with an overlapping halo of shared input voxels; because each projected position depends only on neighbours within a fixed radius, points in the shared region land in the same place in both cubes, and the two cubes therefore present matching geometry to whatever composes them. Finally, we trim the cloud so that the open mesh boundary falls inside the cube face, and weld points lying within 0.25 voxels of one another.

We then assign the point cloud a globally consistent normal orientation using the method of Hoppe et al. [1992], so that all



(a) MLS-collapsed points (b) Guarded ball pivoting (c) Interior loops filled (blue) (d) CVT/RVD remesh

Fig. 3. The per-cube extraction stages on one PHerc0139 sheet. (a) Voxel-centre points after the iterated MLS projection has collapsed the several-voxel-thick prediction to a single sheet. (b) The consistently oriented, guarded ball-pivoting reconstruction; the remaining perforations are prediction gaps, not meshing failures. (c) Clean interior loops are closed (filled regions in blue) while the genuine sheet boundary is left open. (d) The level-of-detail stage on the same sheet: the CVT/RVD remeshing of Section 5.5 relocates samples into a near-isotropic blue-noise distribution.

papyrus surfaces are consistently oriented as the rest of the pipeline requires.

5.3 Guarded Ball Pivoting

Meshing the point cloud requires an algorithm that is local (so cubes can be joined afterwards), fast, and — critically — not designed to output a closed watertight surface. We therefore use the ball-pivoting algorithm [Bernardini et al. 1999] (BPA): three points form a triangle if a ball of user-specified radius ρ touches them without containing any other point; starting from a seed triangle, the ball pivots around each edge until it touches another point, forming the next triangle, and the process continues until all reachable edges have been tried, then restarts from a new seed. This satisfies our criteria at the cost of some small topological artifacts that are straightforwardly repaired (Section 5.4). It also composes: cube surfaces can be joined by setting the cube boundaries to the advancing front at start-up, and it is very well behaved on open surfaces.

The unguarded algorithm, however, has exactly the failure mode this problem cannot tolerate: given two sheets 7 voxels apart and a ball of radius 1.2, a front that reaches the edge of its own sheet will happily roll down onto the neighbour. Our implementation therefore carries five guards, each aimed at a specific mechanism.

- A **dihedral fold-back guard** prevents the front from folding back over itself.
- A **face-coherence guard** rejects any candidate triangle whose normal turns too sharply away from the triangle owning the pivot edge. Without it, the ball tends to roll onto an adjacent wrap instead of across a thin gap in its own sheet, stitching the two layers together.
- An **anti-parallel guard** refuses a triangle that would bridge two oppositely oriented surfaces lying close together, as occurs wherever neighbouring wraps nearly touch.
- A **wall guard** rejects candidates that are simultaneously steep against the local MLS normal *and* long. A fusion bridge stands nearly perpendicular to the sheet it fuses and must reach across the inter-wrap gap, whereas a genuine surface triangle lies parallel to its vertex normals; the conjunction of the two conditions fires on through-thickness bridges and spares sharp folds.
- A **winding growth guard** addresses the gradual failure. Because a front can accumulate error slowly — every individual pivot innocent while the front climbs a through-thickness wall and gains whole turns — each seed carries the branch-cut-free integrated winding phase of Equation (1) along its growth, and a pivot whose accumulated phase leaves a tolerance band is rejected. The surface beyond it is re-seeded as a distinct component rather than silently attached.

Guard thresholds appear in Appendix C, each with the measurement that fixed it.

Adaptive radius selection. A smaller pivot radius can sometimes avoid a local fusion, but it can also discard surface coverage and explode the number of open boundary edges, so ρ is not tuned by appearance. The controller begins at the nominal radius and tries smaller radii only after a winding failure. A candidate is accepted when it contains no abrupt full-turn edges or broad branch-local fusions, retains at least 95% of the baseline point coverage, and introduces no more than 1.25× the baseline boundary edges — a three-way trade between fusion, coverage, and boundary complexity that a single radius cannot resolve for every cube. If no candidate satisfies all three, the component is set aside for cutting rather than meshed at a radius known to fuse it.

5.4 Topological Cleanup

In brief: four repairs, in a fixed order, each removing a defect that would otherwise be indistinguishable from real structure later on.

Separate, then re-project. A single voxel connected-component can hold several *disconnected* surface sheets that were projected and meshed together, because the MLS projection operates on all of a component's points at once and nearby sheets pull on one another.

Before anything else, we separate each meshed component into its connected pieces and re-project and re-mesh each one in isolation — this time from its *original* voxel-centre points rather than from the already-projected vertices, so that the projection kernel sees only same-sheet neighbours and each sheet lands cleanly on its own surface. Re-projecting from the original points proves to be a reusable primitive; we apply it after every split below, to clean the ragged boundary a cut otherwise leaves behind.

Sever fused components. Multiple components may still be erroneously attached within one cube, either through proximity during ball pivoting or, more commonly, through errors in the input prediction. Now that we have meshes, we can detect these connections: we find every connected mesh component in the cube and cast rays from each triangle centroid along its normal in both directions to determine whether the geometry should be split.

We categorize splits into two cases. Where one small bridge connects two levels of geometry — commonly caused by segmenter artifacts (Fig. 4a,b) — we formulate the split as a min-cut problem solved by Edmonds–Karp. For more complex overlapping geometries we use a lifted multicut solver [Keuper et al. 2015] on an overlap-reduction problem in the formulation of Limper et al. [Limper et al. 2018]. After splitting, each resulting piece is re-projected and re-meshed from its parent's original voxel points, so the new sheet boundaries come out clean rather than carrying the jagged trace of the cut.

Close interior holes. After each separation we split pinched vertex fans until every component is vertex-manifold. Holes detected inside a component are then patched (Fig. 4c,d): exact three-edge loops are closed directly, and other clean interior loops are projected to a local plane, triangulated by constrained Delaunay, and lifted back with Liepa-style refinement [Liepa 2003]. Self-intersecting, non-manifold, or clearly exterior loops are rejected rather than force-filled — the distinction matters because sealing a genuine torn edge would convert a known boundary into fabricated papyrus.

Open accidental handles. Ball pivoting can also fuse a sheet to itself *without* disconnecting it: where the advancing front rolls across a thin gap, it can close a loop that raises the sheet's genus into a small handle. Such handles are *non-separating*, so the cut-based splitters above, which sever *separating* necks, cannot remove them. After hole filling we therefore apply a tree–cotree decomposition and open every non-separating loop shorter than a fixed length bound by manifold-preserving vertex duplication. A per-cube sheet has no legitimate scroll-scale handle, so this reduces accidental local genus while leaving the long cycles that represent genuine structure untouched.

5.5 CVT/RVD Remeshing

Earlier versions of ScrollFiesta used quadric error metrics (QEM) to downsample the ball-pivoting output. In practice this created numerous sliver triangles and an uneven distribution of samples, both of which are poison for a later metric parameterization. We therefore perform level-of-detail with a centroidal Voronoi tessellation restricted to the reconstructed surface. We seed interior sites by area and boundary sites along every retained loop, then iterate the variational update of CWF [Xu et al. 2024]: each interior site solves a small fixed-cell system combining the ordinary centroidal (CVT) term with a normal-anisotropy term that pulls samples onto creases and thin features, with the CVT weight decaying across iterations (Appendix C.6). Each updated site is projected back to the source surface, and the output triangles are the restricted-Delaunay dual of the restricted Voronoi diagram (RVD). Uniform density is used per cube at present. Unlike collapse-only QEM, this relocates samples into a near-isotropic distribution, reduces slivers, and does not preserve a poor advancing-front tessellation merely because it arrived first.

One caution is worth recording, because it is a case of the failure described in Section 2.3. A remesher is a purely geometric operator and has no notion of wrap identity: run with an interior edge length above the local pitch, it will happily merge turn N into turn $N{+}1$, producing a perfectly clean mesh that is physically wrong. The CVT stage therefore runs under the same winding bound as the mesher, armed from the measured umbilicus and pitch, and its edge budget is bounded below the clearance rather than chosen for triangle count.

Any component above the current size cap, an empty or invalid dual, or a result failing the manifold checks falls back to the repaired QEM path, which can also be selected explicitly; that fallback rejects edge collapses or maintenance flips that would create an existing edge or reverse winding. Once remeshed, components with negligible area are removed as debris, and halo output is cut at the owned cube planes so that every geometric region has exactly one owner at assembly time.

6 Global Cube Welding

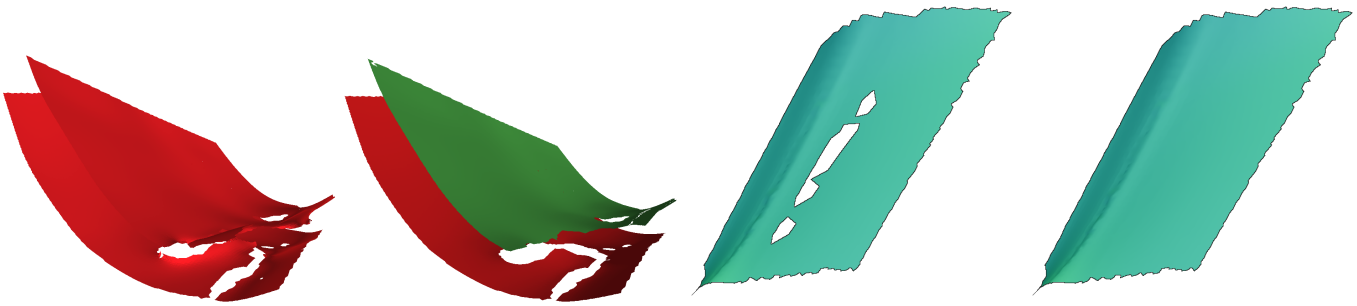
In brief: cubes are meshed independently, so the natural next step is to sew their sheets back together across the cube faces. ScrollFiesta can do this, and the result is instructive — but the active whole-sheet path deliberately does not require it, for a reason worth stating carefully.

Welding is optional in the current system, and understanding why is more useful than the weld itself. A geometric weld makes an assertion of the strongest possible kind: these two triangles are the same surface. Once made, that assertion is indistinguishable from a triangle that was measured, and by Section 2.3 no downstream geometric test can question it. Meanwhile, per-cube reconstruction and remeshing can leave divergent seam triangulations even where the halo voxels agree, so the weld is being asked to make a hard topological claim on the basis of a soft geometric coincidence. The ribbon fit of Section 8 can instead close cube seams from material claims, treating continuity as a *fitted relation* that carries its own confidence. The active path therefore consumes the pre-weld pile.

We retain the weld for three reasons: local welding is genuinely useful for verified patches; the welder is used as a sub-engine by the unwrapper to ask whether two sheets adjacent in (u, v) are actually neighbours in the scroll; and it is the correct frozen yardstick for upstream A/B comparisons and for comparison with earlier published results.

6.1 Seam-Band Reconstruction

We first place every accepted cube mesh in volume coordinates and concatenate it under the ownership established by halo trimming.



(a) Fused pair, before (b) Severed at the neck (c) Perforated sheet, before (d) Interior loops closed

Fig. 4. The two mesh-space repairs that must happen before any parameterization is attempted, on real PHerc0139 cubes. (a,b) The U-Net has fused two sheets through a thin neck of bridge material; the bridge is detected and severed by min-cut, recovering two distinct sheets. A fusion left in place would weld two wraps into one and make the winding topology unrecoverable — and, per Section 2.3, nothing further downstream could tell. (c,d) Holes left by the ball-pivoting front and by gaps in the prediction are closed: exact three-edge loops directly, larger clean interior loops by constrained Delaunay with Liepa-style refinement [Liepa 2003]. Self-intersecting, non-manifold, or clearly exterior loops are rejected rather than force-filled, so a genuine sheet boundary is never sealed shut.

The uniformly coarse CVT interiors are desirable for scale, but their triangles are too large for the weld's ball — radius at most three voxels — to initialize reliably. We therefore detect every internal cube plane and refine only a narrow seam band to approximately 3.5-voxel triangles (Fig. 5). Boundary half-edges in that band seed the same BPA used per cube, and four sub-processes then connect matching fronts while rejecting invalid joins.

A **sliver cull** first removes near-degenerate boundary triangles that would prime the bridge ball off a needle. The **bridge front** escalates its radius adaptively from the local boundary spacing, capped so that twice the maximum radius stays below the measured ∼7-voxel inter-wrap clearance: welding two wraps together is thereby made geometrically impossible for the bridge, rather than merely penalized. This is a deliberate design choice — where a physical constant of the object is available, we prefer to make the failure unreachable rather than expensive.

Every candidate bridge face must additionally pass a **winding test**. Using the winding coordinate of Equation (1), the phase gap w between front and candidate must stay within a soft tolerance of 0.35 turns, with a 0.40-turn hard cap that separates legitimate grazing-seam closures from cross-layer and next-wrap stitches. The same test is what keeps recto from welding to verso. Finally, a **sheet-correspondence pass** recovers legitimate same-sheet closures that the conservative test rejected: it confirms a mutual-best one-to-one sheet pairing across each seam from bridge votes and boundary overlap, then re-welds only that confirmed pair, on a point cloud restricted to those two sheets.

6.2 Recoarsening Under Invariants

After each weld the temporary fine band is recoarsened and optimized with a *pinned-boundary* CVT/RVD pass. Boundary positions are held bit-exact while interior sites solve the same centroidal and normal-anisotropy objective as Section 5.5. A proposed band patch is accepted only if every pin is represented, the complete boundary-edge set is unchanged, all edges remain manifold, and both component count and Euler characteristic equal their pre-remesh values. Tiled passes with an offset allow long seams to be processed at bounded memory, and if any invariant fails, that tile simply retains the pre-CVT weld. The pattern here recurs throughout the system: an optimization is allowed to improve a representation only when it can prove it did not change what the representation means.

6.3 Checking the Result

Orientation consistency alone is insufficient to guarantee a valid scroll: a perfectly manifold bridge may still connect turns that are far apart in scroll phase. The global scan therefore measures abrupt full-turn edges *and* computes a branch-local phase span, which is what detects broad fusions assembled from individually plausible edges. A phase sever can remove faces crossing those discontinuities, but its output is not relied on until every resulting component also satisfies the broad-span test.

The flat weld remains available for a group of placed cubes. The complete demonstration here is a flat $4 \times 5 \times 5$ block; the August $4 \times 21 \times 21$ experiment was decomposed into axial strips at the atlas level. The bottom-up hierarchical runner has been removed from the public release. Native sheet assembly instead places and repairs charts directly from the per-cube mesh pile (Section 10).

7 Sheet Books: Identity and Reading Order

In brief: this stage answers two questions and deliberately leaves a third alone. Which pieces of the prediction are one sheet? In what order are those sheets read as one unrolls the scroll? It does not *answer where each sheet lands on the page — that is Section 8 — and, crucially, it never glues two sheets together in order to make its own answer tidier.*

The natural temptation, once one has a mesh, is to treat each connected component as the unit of identity and assign it an integer wrap. That is what a previous version of this system did, and Section 12.5 reports how it failed. The difficulty is that mesh connectivity is a statement about geometry, not about papyrus: two pieces of one sheet separated by a hole are disconnected, and two

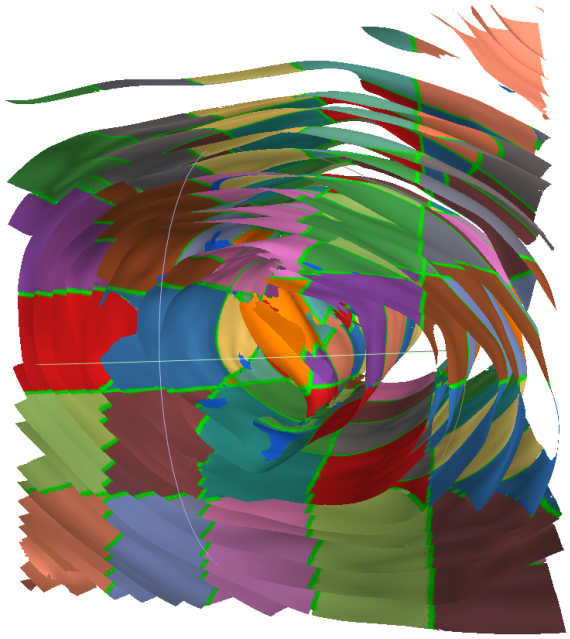


Fig. 5. The welded 4 × 5 × 5 block with each cube's sheet coloured independently and the refined seam bands drawn in green. The weld bridges the ∼1-voxel trim gap between adjacent cubes only inside these bands; the winding test and the capped bridge radius decide which fronts may join, so a sheet continues across a seam exactly when the two cubes agree it is the same wrap.

different wraps joined by one fusion are connected. Identity has to be established from evidence that can distinguish the two.

We therefore separate three questions that were previously entangled. Which prediction cells constitute one locally coherent sheet? What winding relation is supported between two such sheets? And how should the accepted geometry be parameterized metrically? A *sheet book* answers the first two without fusing components. A *quad ribbon* answers the third without revising their physical identity.

7.1 From Slice Skeletons to Sheets

The tracer processes the completed prediction as axial shards of full in-plane slices. On each slice it extracts a pruned skeleton, decomposes it into branch-safe curves, and associates curves between neighbouring planes into *material tracks*.

The in-plane fusion cut. A single traced curve is not automatically one sheet. A path may follow one wrap, cross a prediction fusion, and continue along another wrap while remaining, throughout, one connected curve — and every geometric property of that curve is unremarkable. The fusion cut detects the case *along the curve* rather than at a point: samples are classified by whether they lie closer to their own bootstrap track or to another track, and a sustained run of foreign ownership bounded by sustained runs of own ownership triggers a cut. Requiring sustained runs on both sides is what distinguishes a genuine wrap hop from noise at a contact. Association then restarts on the split pieces, so no downstream index ever inherits the fused identity.

Lifting and checking. For each candidate track, sheet lifting fits a dense single-valued 3-D field over its supported domain and records *direct support* separately from *interpolation* — a distinction maintained from here to the final pixel.

The ordering of the check is the point. Geometry and support masks are frozen *before* any RAW is sampled; only then is held-out CT consulted, and it must correlate better than a null placement. A track that is too fragmented, shadowed by another surface, normal-flipped, or placed over black CT is rejected. Freezing first is what makes the CT test meaningful: a fit allowed to see RAW would have optimized against exactly the quantity later used to judge it.

The check catches real failures at both extremes — local bad interpolation, and large prediction phantoms. The $4\times21\times21$ run rejected a 400,000-cell phantom whose fitted geometry landed entirely where the CT was zero; without the check, it would have improved every coverage statistic in the paper.

Table 3 shows the ladder after the in-plane fusion cut. Every rung carries more than 70% of evidenced material, and the shape of the numbers is itself informative: at the largest rung, 6,188 of 6,500 lifted sheets pass, yet the single largest sheet carries only 0.15% of the material and reaching 70% of it requires 5,493 tracks. The evidence at full scale is genuinely fragmented. We record that rather than concealing it behind a monolithic fused surface, because a fused surface would have had to invent precisely the connections the fragmentation reports as missing.

7.2 Winding Assembly Without Geometric Fusion

In brief: accepted sheets still need a reading order — which is the outer neighbour of which. We measure that from how far apart two sheets sit radially, in units of the local *spacing, and accept a relation only when it is close to a whole number of turns.*

This section reports book-level order on traced prediction tracks. Section 8.1 addresses the complementary mesh-level problem: fragmented ball-pivoted components receive generalized-winding unary evidence and are optimized jointly as an MRF. The latter does not replace the scale ladder below, and the ladder does not test the new absolute cue.

For two tracks that coexist on sampled planes, we measure their signed separation along the local umbilicus ray and normalize it by the pair's local median inter-wrap spacing. An edge is admitted only when the resulting offset lies within 0.25 turns of an integer; ambiguous offsets contribute no edge.

Using *local* spacing is essential rather than a refinement. Adopting one global 9.5-voxel pitch mislabelled physically separated 12–16-voxel delaminations as two turns rather than one — the sheet had split, widening the local gap, and a global constant read that widening as an extra wrap. Pitch is a property of a neighbourhood, and treating it as a constant of the scroll manufactures winding errors in exactly the damaged regions where they are hardest to notice.

Composing the relations. The accepted integer relations form a weighted graph, and it will generally contain cycles that do not close.

Table 3. Sheet-book ladder. "Passing material" is the fraction of evidenced prediction cells carried by sheets whose geometry, support, interpolation, and held-out RAW checks all hold. The criterion is material-weighted rather than count-weighted, because many tiny fragments may fail while the book still carries most of the evidenced papyrus, and many trivial sheets may pass while a major sheet fails.

Grid	Sheets lifted	Sheets passed	Passing material	Main ordered span	Rejected / order suspended
$4 \times 5 \times 5$	50	49	83.7%	76 tracks / 17 winds	1 sheet; 0 ordered tracks
$10 \times 10 \times 10$	400	385	74.3%	1,058 tracks / 47 winds	15 sheets; 0 ordered tracks
$4 \times 21 \times 21$	1,150	1,104	71.9%	2,930 tracks / 102 winds	46 sheets; 51 tracks
$21 \times 21 \times 21$	6,500	6,188	71.4%	15,457 tracks / 112 winds	312 sheets; 174 tracks

A Kruskal-style *maximum-support forest* establishes the initial frame, so that when a cycle disagrees, responsibility falls on its weakest-supported edge rather than on an arbitrary one.

A forest alone is not enough, because a single wrong forest edge displaces an entire subtree by a constant offset and no local check can see it: every relation inside the subtree remains perfectly satisfied. We therefore evaluate small integer *subtree shifts*, ranked by the conflicted evidence mass crossing each forest cut, and accept a shift only on a strict reduction in crossing conflict weight. This is loop closure: the cycle-closing relations that the forest discarded are precisely the ones that can detect a globally displaced subtree, and using them is what turns a spanning structure into a checked one.

Quarantine. If a few tracks remain inconsistent with most of their neighbourhood, we suspend those tracks' *order relations* while retaining their geometry. We do not discard the geometry, and we do not dissolve the component. The reasoning is simple: a contradictory triangle of relations cannot be made consistent by silently deleting one edge and calling the remainder an order. Quarantine is capped at 5% of a component, and it removes only a claim about reading position — never papyrus.

Results. The ordered fraction is 100% on $4 \times 5 \times 5$ and $10 \times 10 \times 10$, 98.9% on $4 \times 21 \times 21$, and 99.3% of assigned material on $21 \times 21 \times 21$. On the largest rung, collection covers 95.9 million sampled points; 23,900 accepted winding edges arise from 154,545 candidates, 25 subtree repairs are applied, and 174 tracks (1.1% of the ordered component) have their order suspended. The median snap residual is 0.12 turns, comfortably inside the 0.25-turn admission bound. Geometry remains separate throughout: a winding edge states an order relation and never welds two tracks.

The result is an ordered book, not yet a complete solid page. Radial order is established because it is measurable — a signed separation against a local pitch. Same-wrap continuation across a large unsupported arc has no comparable local measurement, and until it does, a missing track remains a hole in Fig. 6.

8 The Metric Quad Ribbon

In brief: this is the flattening. Given accepted geometry, we must give every point a position on the page such that horizontal distance is true distance along the unrolled papyrus. Get this wrong by a few percent and the drift accumulates over metres of sheet until letters no longer line up with themselves; get the winding wrong and two different turns land on top of each other.

8.1 Probabilistic Winding Inference

Each disconnected mesh component has a continuously lifted cylindrical phase q_i, but that phase is known only up to an integer turn k_c. A maximum-support forest gives a coherent initial gauge, yet a forest necessarily discards its cycle-closing relations: one wrong heavy edge can shift an entire subtree while every relation inside that subtree remains satisfied. The current winding stage therefore treats the forest as an initializer rather than as truth.

A probabilistic field observation. For an oriented triangle soup, let

$$g(x) = \frac{1}{4\pi} \sum_f \Omega_f(x)$$

be its generalized winding number, with Ω_f the signed solid angle of face f [Jacobson et al. 2013]. Components are first orientation-voted against the umbilicus. At every sampled vertex v_i we evaluate the field on both sides of the sheet,

$$m_i = \tfrac{1}{2}\big(g(v_i + \epsilon n_i) + g(v_i - \epsilon n_i)\big), \qquad j_i = g(v_i - \epsilon n_i) - g(v_i + \epsilon n_i).$$

A clean, consistently oriented sheet has $j_i \simeq 1$. A duplicate same-orientation shell tends toward two, an orientation cancellation toward zero. We therefore weight a sample by $r_i = \exp[-\frac{1}{2}((j_i - 1)/0.35)^2]$ and discard $r_i < 0.05$. The implementation chooses the exact boundary-edge reduction of Xie et al. [2026] when its measured work is affordable and otherwise uses the second-order fast expansion of Barill et al. [2018], with exact triangles in the near field.

For component c, the field votes for its missing integer through the weighted median of $-m_i - q_i$. Its uncertainty is the larger of one turn and 1.4826 times the weighted median absolute deviation; the one-turn epistemic floor is intentional, because an incomplete scroll field can look locally sharp while compressing the global range. Within each measured relation island, a positively sloped robust affine calibration maps the field cue into the forest gauge. A fit with fewer than three supported components, slope outside $[1/8, 8]$, or $R^2 < 0.25$ contributes no unary.

MRF reconciliation. Let k_c be the integer correction for component c. Every admitted continuation or radial-order observation remains an individual pairwise factor with target t_e; contradictory factors

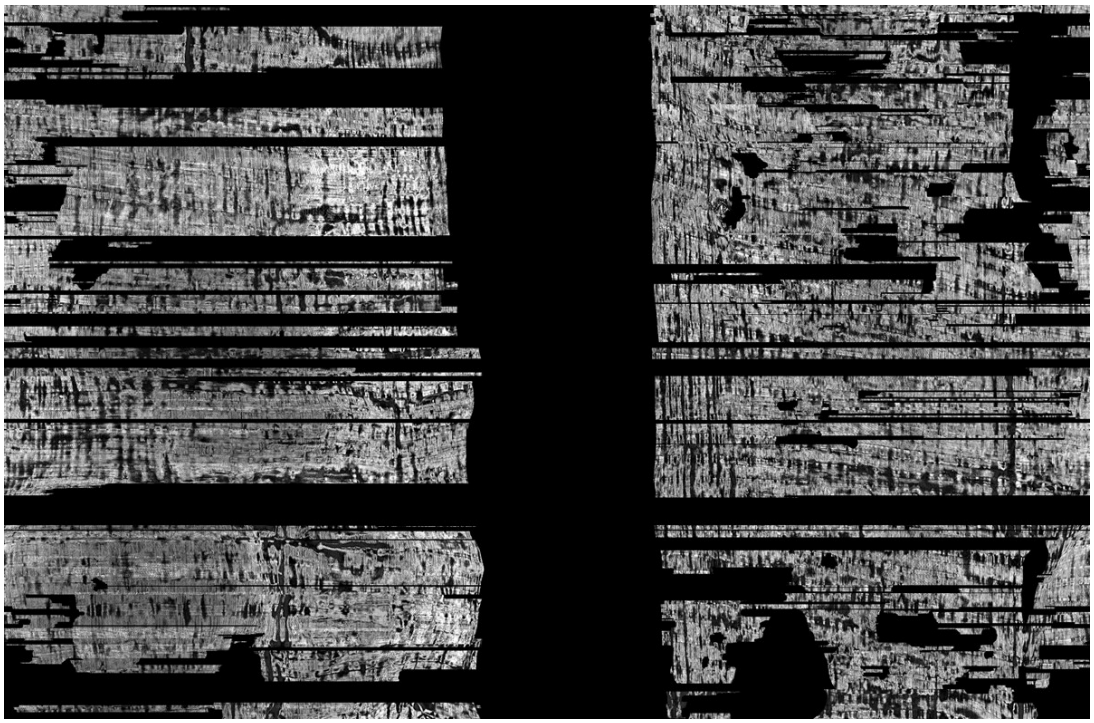


Fig. 6. One 21 × 21 × 21 winding page assembled from the sheet book. Horizontal fragments are texture-bearing track contributions placed in the inferred reading frame; black gutters and holes remain wherever the book has no accepted track. This is one winding view, and deliberately not a claim of complete page coverage — the visible gaps are the fragmentation reported in Table 3, shown rather than filled.

are not averaged away. The optimized energy is

$$E(k) = \sum_c \alpha_c \, \rho_{2.5}\left(\frac{k_c - \mu_c}{\sigma_c}\right) + \sum_{e=(a,b)} \beta_e \, |(k_b - k_a) - t_e| + \sum_{(c,\ell)\in\mathcal{P}} \gamma_{c\ell} \, [k_c = \ell], \tag{2}$$

where ρ is Huber, $(\mu_c, \sigma_c, \alpha_c)$ come from the calibrated field, and $\mathcal{P}$ is an optional sparse set of forbidden assignments. Continuations retain a modest preference over order factors, but the historical forest-priority offset is removed because it encoded an ordering artifact rather than evidence mass. Alpha–beta swap graph cuts [Boykov et al. 2001] optimize residual labels in $[-2, 2]$ around the current state; after a nonzero update the state is recentered and solved again, for at most four hierarchical rounds. The relation-island root is fixed. Output confidence is the sharpness of the local conditional label distribution with neighbouring MAP labels held fixed, not a claimed exact global marginal; sharpness below 0.35 is reported as unresolved.

Conflict penalties. A second pass asks whether the current labels place spatially remote geometry in the same developed cell. Claims are keyed by relation island, axial bin, absolute integer turn, and wrapped-phase bin. Samples within 0.55 local pitch are one physical claimant; only clusters separated by at least eight pitches and 48 voxels are mutually exclusive. The current test deliberately ignores ordinary two- and three-layer occupancy, which belongs to quadribbon layer handling, and targets only a cell with at least four claimant clusters. The claimant supported by the greatest mass of satisfied relations keeps the cell. A component may receive a forbidden (c, k_c) assignment only if it loses in at least one such cell and wins in none. During the outer solve, every component not currently occupying a forbidden assignment is fixed, so the losing batch can use relations to the unchanged field without dragging that field with it. A band of eight turns around the base MRF prevents an integer cascade.

This outer pass does not currently work, and the way it fails is informative. The first unguarded experiment expanded the $4 \times 5 \times 5$ winding span from 14.13 to 32.41 turns over 24 exclusion rounds: forbidding an assignment is cheapest to satisfy by moving a component a whole turn away, so the penalties resolved collisions by inventing turns. Restricting them does not repair the underlying tension. A broad sweep counts 21,420 conflicting cells; winner- and context-lock variants reduce this to 11,099/10,214 while driving continuation/order satisfaction to 0.95789/0.66605 and 0.95580/0.80329 — the exclusions are paid for out of the relations that produced the labels. Restricting attention to the four-claimant case leaves two cells and four losing claims; the one-bin variant clears both in four rounds using 20 forbidden assignments and preserves the 14.13-turn span, but creates 42 residual relation conflicts and raises MRF energy from 200.486 to 1,119.330, while requiring the loss in two cells

makes no exclusion at all. A count of spatial collisions is simply a weaker measurement than a relation, and every configuration we tried either leaves the conflict or spends more relation evidence than the conflict is worth.

Standalone artifact. `scroll_ribbon -winding-only` stops before any ribbon graph or metric solve and writes topology-identical VMESH geometry with winding U, axial V, plus field, jump, label, confidence, and JSON sidecars. This separates the discrete labelling from everything downstream that could make a wrong label look visually smooth.

8.2 From Mesh Pile to Ribbon

The active executable takes a recursive pile of per-cube meshes and writes one staged result:

mesh soup → fit and metric solve → untangle
→ metric re-solve → layer bakes and composite.

Global geometric welding is omitted, and the feasibility of omitting it was measured before the path was built. On the $4 \times 5 \times 5$ pile, 100 per-cube meshes contain 783 raw geometry components; claims ownership and phase-continuation association reduce these to approximately 25 material candidates with saturated cube-seam continuation. The welded mesh, for comparison, yields 24. Cross-cube continuity is therefore obtainable as a fitted relation, and does not require two independently remeshed seam triangulations to agree as geometry.

Rows from slices. The fit intersects the mesh soup with planes normal to the scroll axis. Mesh-edge identity chains the resulting intersections into *row observations* — chaining by shared edges rather than by endpoint proximity, since proximity is exactly the unreliable quantity here. Winding registration supplies continuously lifted cylindrical phase for those rows.

Continuation, and its limit. Local continuation can close short discontinuities but must not span an entirely missing arc. Disjoint phase intervals are joined only when three conditions hold together: their angular gap is at most 1.3 radians, their axial supports overlap, and their radial change matches the local spiral pitch (within half a pitch). The third condition is the one doing the work — it asks whether the junction *continues the wrap* — and without it, the first two would happily join a sheet to its own next turn. Gaps of 4.2 radians or more stay split for an evidence-backed pass. Branch lanes and delaminations are retained as parallel material candidates rather than averaged, since averaging two plies destroys both.

This continuation rule has a measured origin. Before it existed, one physical sheet tiled consecutive angular intervals as separate material islands: a chain of intervals with gaps of 0.36–1.19 radians across holes of 175–525 voxels, each individually plausible as a distinct island. The sheet was one sheet; the fit had simply had no vocabulary for saying so.

8.3 Metric Projection

Two experiments isolate complementary halves of metric parameterization, and we keep them separate deliberately: one establishes what a carried frame already determines, and the other establishes what independent observations can add.

Fixed-topology projection: no solver at all. In the fixed-topology campaign, each row chain is oriented against the carried frame, assigned exact 3-D arclength s, and projected as

$$u_i = F(s_i + g_c), \qquad g_c = \mathrm{median}_{i \in c,\ |du/ds - 1| \le 0.15}\,(u_i^{\mathrm{seed}} - s_i),$$

where F is a single reciprocal-integrated metric-rate map shared by the entire carried axis, and the per-chain offset g_c is a robust median: only samples already locally metric to within 15% vote on where their chain sits. Detached islands are placed rigidly through the monotone authoritative phase-to-u map of the dominant component. No recomputed principal-angle fallback may place an unanchored island, because a fallback that guesses is indistinguishable in the output from a measurement that knows.

On $4 \times 5 \times 5$ this path needs no numerical solver, preserves all 4,033,291 faces, and runs in 9.6 seconds. Its value is as a control: it proves that exact metric projection plus authoritative phase placement — with no optimization anywhere — already recovers every user-marked dropout that another evidenced layer covers.

Robust consistency: where independent observations disagree. The pre-weld fit contains real cracks and contacts that a fixed carried topology does not. Here we solve their consistency explicitly rather than accumulating local corrections, which is the difference between reconciling measurements and propagating an error. For scalar u values x, rows encode observed differences d_i:

$$\min_x \sum_{i \in \mathcal{R}} w_i\, \rho_{c_i}\big((x_{a_i} - x_{b_i}) - d_i\big),$$

where ρ is Huber. *Metric rows* ask consecutive samples to advance by their measured arclength; *vertical and face-contact rows* ask for zero drift; and *crack ties* close an evidenced fissure — admitted only where neither facing side already carries a face tie, so a crack cannot be closed twice by two different mechanisms. One pin per component fixes each null space.

Normal equations are assembled into lower-triangular sparse form and solved by TAUCS supernodal multifrontal Cholesky, with at most three IRLS rounds recomputing exact residual weights. The $4 \times 5 \times 5$ system has 2.04 million unknowns and 12.2 million nonzeros and factors in 8–11 seconds. Its 28,597 crack ties reduce visible ±20–25-pixel slips to approximately ±1.3; texture cross-correlation measures a median sheet-seam shift of 3 pixels rather than 10–25. Island placement is then repeated against the solved, coherent u field before a global non-negative re-zero.

Running island placement twice is not redundancy. The phase-to-u map used on the first pass is built from the carried u, which at the inner turns is compressed and locally non-monotone; placements derived from it collapse onto a plateau. Rebuilding the map from the *solved* field and re-placing is what recovers inner-turn order.

Neither solver owns identity. This is the invariant that makes the two experiments composable. Solver-free projection establishes what the carried metric frame already determines; IRLS is used only where independent pre-weld observations supply residual constraints. Neither chooses winding identity. That authority stays with

continuous phase and the assembly graph, which is what prevents a least-squares residual from purchasing a wrap change.

8.4 Reading the Result

The final image is also the strongest available test of the parameterization, and it is a physical test rather than a numerical one. For each texel of the developed (u, v) image we sample CT intensity at the corresponding surface point. If u is a correct developed arc length, the two papyrus fibre families remain continuous across the page; a wrong parameterization shears, duplicates, or breaks them. Dark bands that persist coherently across rows are cracks and damage rather than parameterization artifacts. On synthetic spirals with known geometry, the local solver tracks analytic arc length to within a fraction of a percent across all turns.

The $10 \times 10 \times 10$ region (Fig. 8) is currently unwrapped by composing the two campaigns directly: the solid connected quad ribbon supplies topology and a first-guess parameterization, and the TAUCS consistency solve then re-fits u for constant arclength across v on that fixed topology — 16.4 million unknowns, 98.7 million nonzeros, three IRLS rounds, 25 minutes. The re-fit removes the inter-wrap drift responsible for earlier waved composites: contact residuals collapse from $p10/p90 = -1304/+324$ pixels to $-2.5/+2.5$, while the row metric is essentially untouched ($|du/ds - 1|$ mean 0.058 before and after). That combination — a hundredfold improvement in cross-wrap consistency at no cost in along-row metric — is the signature of a drift that was a frame error rather than a measurement error. The bake samples the complete masked CT volume; for context, the original run of this ribbon was missing 728 of 1,728 RAW cubes.

8.5 Integration State

The consolidated command is `quadribbon <mesh_dir> <out_dir>`. It writes the concatenated soup, fitted ribbon, first metric solution, untangled geometry, second metric solution, layer manifests, RAW bakes, composite, label image, and per-stage logs. Stage artifacts are resumable, and production constants — axis, pitch, CT source, bake window, dark threshold — come from one frozen configuration rather than from tuning flags on the command line.

At the current snapshot, all module self-tests pass and the canonical $4 \times 5 \times 5$ run from the pre-weld pile completes end to end. The fixed-topology and solid-ribbon campaigns reported in this section and the next are the quantitative evidence for parameterization, attachment, and compositing; we do not relabel them as output of the integrated command. Larger canonical pre-weld piles have not yet traversed this exact sequence.

9 Solidification, Untangling, and CT Attachment

In brief: parameterization makes defects measurable, but measurable is not the same as fixable. This section covers four operations: filling holes inside one accepted sheet; separating a surface that passes through itself; nudging geometry onto the papyrus the CT actually shows; and choosing which layer supplies each output pixel. Each accepts a change only when the geometry it produces measures better than the geometry it replaces.

The temptation at this stage is a familiar one. Once the sheet is flat, every defect is visible and every defect looks fixable: a hole can be interpolated, a dark island can be moved to the nearest bright voxel, a raster void can be filled. Each of those operations improves the picture. Three of the four can also destroy the result, because each can silently connect material across a wrap boundary. The design here is therefore built around what each operation is forbidden to touch.

9.1 Component-Bounded Solid Fill

A fitted material track is single-valued over most, but not all, of its parametric bounding box. The solid-ribbon builder allocates a regular quad lattice over the union of each component's occupied row and column spans. Directly fitted vertices form Dirichlet data. Missing positions inside that component solve the four-neighbour harmonic problem

$$x_i = \frac{1}{|N(i)|} \sum_{j \in N(i)} x_j, \qquad i \in \mathcal{H}_c,$$

independently for each world-coordinate component, with a sparse direct solve for small holes and preconditioned iteration for larger ones. A cylindrical alternative continues radius and lifted phase under the same boundary conditions, using local pitch to preserve turn order.

The single most important property of this operator is negative: the fill domain $\mathcal{H}_c$ never crosses a component label. Separate tracks, separate wraps, and quarantined order claims remain separate even where filling the rectangle between them would produce a visibly more complete image. A neighbour carrying another component label is simply not in the stencil, and there is no inter-component fill state anywhere in the implementation.

On an early full-grid implementation this operation added 18.8% support on $4\times5\times5$ and 15.8% on $10\times10\times10$, with approximately zero dark pixels and 0.04% multiple coverage in the immediate structured bake. Those aggregate numbers turned out to be insufficient, and the reason is worth recording: generated continuation can look geometrically regular while crossing a wrap that the *source prediction* had already fused. The fill was locally correct and globally wrong, and no measurement of the fill itself could tell. We therefore retain direct and filled labels through every subsequent measurement, and never report full-rectangle quality without a support-masked companion.

9.2 Exact Collision Untangling

In brief: a coiled ribbon fitted from noisy observations will, in places, pass through itself. That is physically impossible, and no amount of CT-guided correction can repair it, because the two interpenetrating layers are competing for the same voxels.

The untangler first applies a sparse turn-order escape, then builds an *exact* triangle-contact set — exact pairwise tests, not raster occupancy, because occupancy at any practical resolution cannot resolve two sheets 7 voxels apart. Each contact identifies two continuous phase branches and their radial order. A scalar radial displacement field is then solved over an active collar with screened parametric smoothness, unilateral inter-turn clearance, and the pre-escape ribbon as a rest shape. Every trial is tested for exact pair intersections, folds, and orientation reversals.

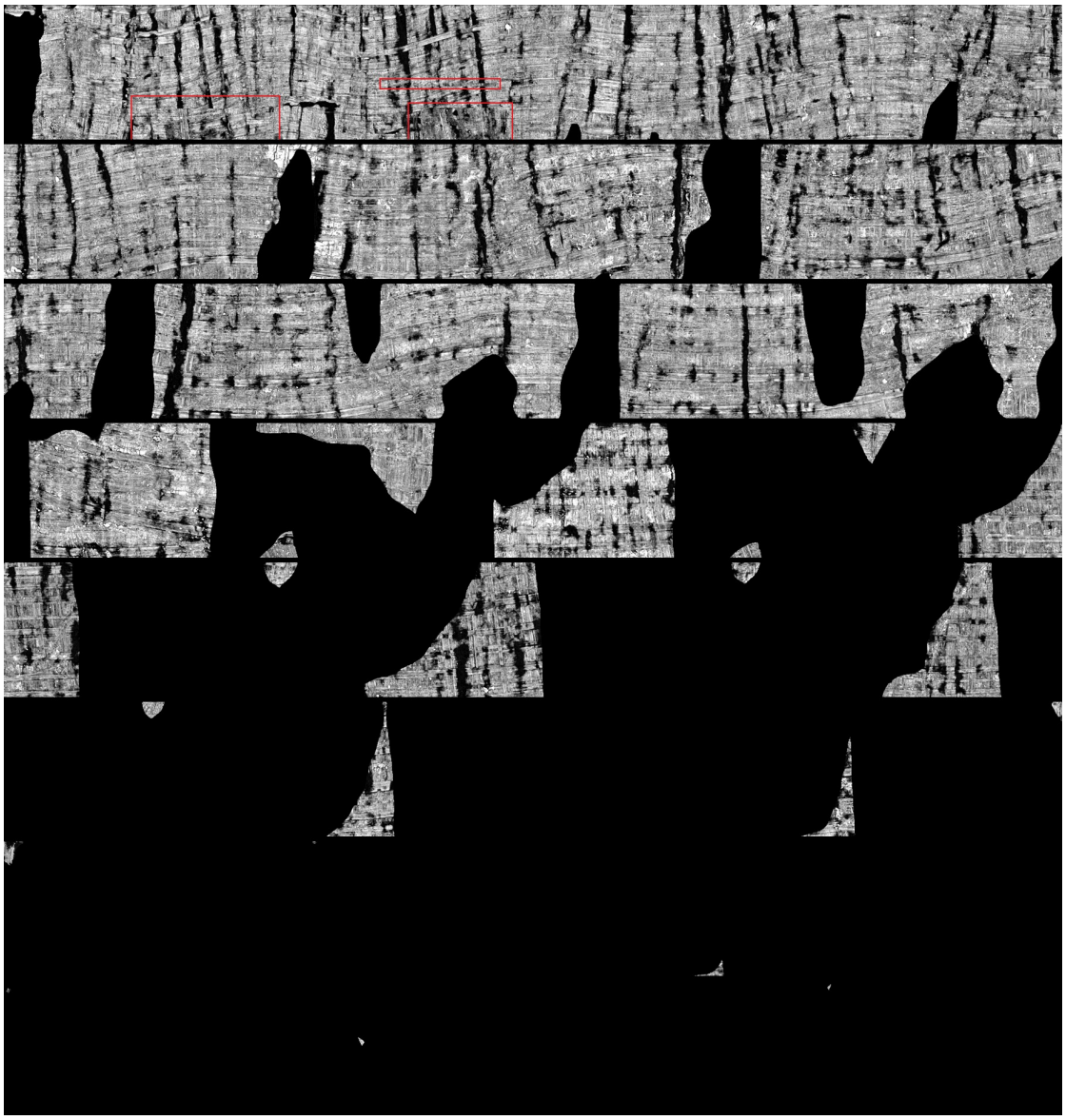


Fig. 7. The $4 \times 5 \times 5$ fixed-topology metric composite, reformatted from its native $31{,}127 \times 508$ ribbon into reading rows. All 4,033,291 faces are conserved. Red boxes mark the previously identified dropout sites; evidenced secondary layers now cover them. Black areas remain unsupported rather than being presented as papyrus.

One scheduling detail is load-bearing. Temporary growth in the contact count is accepted, but only while the number of *phase-order violations* falls. Without that allowance an intersection curve is pinned at whichever triangle pair it first touched and can never travel across the tessellation to a configuration where it can be removed; the measured sequence on the first 15% diagnostic is $428 \rightarrow 2{,}138 \rightarrow 7{,}630 \rightarrow 104 \rightarrow 0$, and a monotone acceptance rule stalls permanently at 428. Once long contacts are cleared, a settle phase pulls the ribbon toward its measured rest coil while enforcing every accumulated contact.

On that diagnostic, sparse repair alone stalls at 428 long contacts and four elastic-shell steps reach zero with no folds. After two guarded CT passes and metric relaxation the crop contains two benign local overlap classifications, zero stabs, and zero folds. A confidence-preserving coarse variant reduces the working mesh to 8.39% of its vertices, lowers the solve from 39 minutes to 2 minutes 14 seconds, clears 3,496 long contacts, and ends at zero overlap, stab, and fold with Sander $p95/p99$ of 1.359/1.461.

The full $4 \times 5 \times 5$ solve takes 7 minutes 29 seconds, against 16.5 non-converged hours for the preceding method. The shell reaches

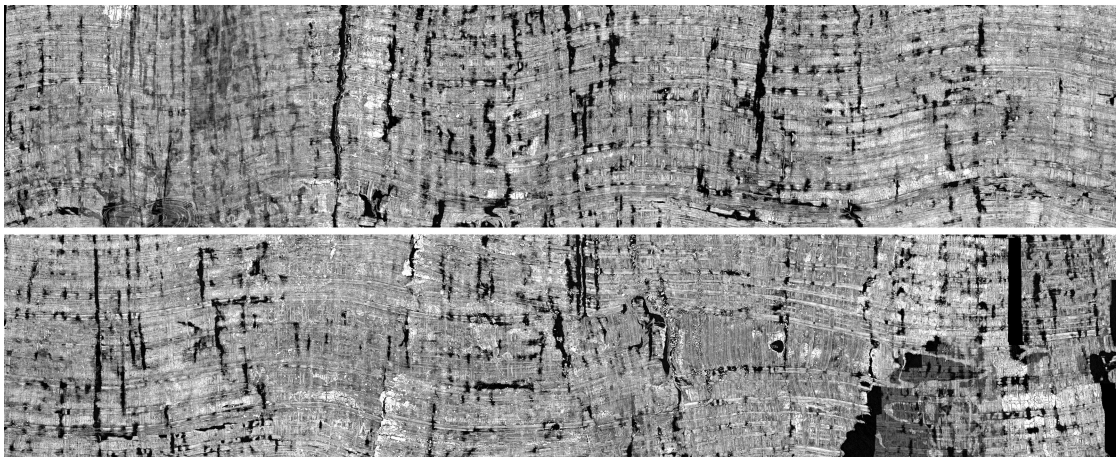


Fig. 8. A 10 × 10 × 10 sheet: the central 12,800-pixel span of the 33,338 × 1,276 composite, reformatted into two reading rows. The solid quad ribbon supplies connected topology; the constant-arclength re-fit supplies the metric (±1300 → ±2.5-pixel contact residuals). Dark regions are cracks and unsupported material, not fill.

its best exactly-tested state at 4,688 contacts rather than the earlier 147,728, and guarded snap reduces this to 4,073. The output has strict RAW coverage 294/294, 100% raster coverage, and no skipped or multiply covered pixels. Its 11.53% dark rate is 0.91 percentage points worse than the less topologically sound predecessor — a real cost of untangling, since a sheet forced to be intersection-free gives up brightness that a self-overlapping sheet had obtained by sampling the same material twice. The shell does not reach zero on the full sheet: its 313,000-contact set becomes mutually restrictive, and contact pruning together with the newer pre-weld input are the next controlled variables.

The 10 × 10 × 10 diagnostic establishes a different and more important boundary. Its fitted, measured region is strong — dark 4.58%, Sander $p95/p99$ 1.13/1.23, only 0.04% at ≥ 4× — but 41% of the full rectangle is generated continuation crossing wraps that were fused in the source prediction, and that continuation is dark 25.7% with 30.6% at ≥ 4×. No untangler can recover an identity that was already destroyed upstream; those pixels remain outside the shipped mask, and the failure is attributed to the prediction rather than to the geometry.

9.3 Transactional CT Attachment

In brief: the fitted surface is a good guess at where the papyrus is, but the CT knows better. Snap moves geometry onto the measured papyrus band — and then has to prove, against six independent measurements, that the move was an improvement.

The first geometry is a warm start, not a set of immovable measurements. Vertices backed by direct fit evidence stay fixed; interpolated vertices, and fitted vertices touching dark emitted triangles, may move. For each active region the snapper searches within a winding-aware corridor for a coherent papyrus-band target, fits a robust vector field from the accepted targets, applies a smooth displacement, remeshes correspondences where needed, and rebakes against the authoritative RAW volume.

Two aspects of target selection deserve comment. The corridor is winding-aware, so a candidate can never be found on the far side of an intervening wrap, however bright that wrap is. And the target is a *band*, an oriented dark-to-bright transition scored for local coherence, rather than an intensity maximum: papyrus placement is not an intensity-maximum problem, and a vertex that chases the brightest nearby voxel will happily walk off its own sheet.

The accept/reject rule. A pass is committed only when it satisfies *all* of the following relative to its input state: no new orientation violation; no growth in the contact set; no multiple coverage and no raster hole; no fixed-window darkness regression outside an explicit Pareto allowance; and no material Sander-distortion regression. A candidate can therefore improve CT intensity — the thing one is most tempted to optimize — and still be rejected.

The loop is iterative and component-selective. The outer loop rebakes the accepted state, detects broad residual black regions, proposes releasing only their suspect evidence, and evaluates each component independently. A rejected component remains bit-identical to its previous state; accepted and rejected components are then rebuilt into one mixed monolith. Correspondence remeshing is likewise objective-selected rather than mandatory. The process stops when no component proposal passes, which makes the terminal state a *measured fixed point* rather than a chosen round count — a distinction that matters, because a hand-selected iteration count is indistinguishable from stopping when the picture looked best.

Result. The 4 × 5 × 5 V83 fixed point contains 4,100,366 unique vertices and 8,168,120 faces. Its independently baked fixed-window texture has 258,276 dark pixels (3.162%), Sander $p95/p99$ 1.190/1.380, 11,142 samples at ≥ 2×, 428 at ≥ 4×, and zero UV skips, 3-D skips, multiple coverage, enclosed holes, or coverage faults. Relative to V67 it removes 768 dark pixels, 323 ≥ 2× samples, and 78 ≥ 4× samples. One further proposal round leaves every previously rejected component byte-identical except one, and that component worsens

darkness and both distortion tails and is rejected — which is what convergence looks like when it is measured rather than declared.

9.4 Raster Correctness, Layering, and Output Masking

Sampling. The rasterizer samples pixel centres. The first exact-metric ribbons exposed a half-texel bounding-box error: enumerating a triangle from $\lceil u_{\min} - 0.001 \rceil$ is correct for historical integer-lattice UVs but drops the first row or column of faces at generic sub-voxel phase. The correct bound is $\lceil u_{\min} - 0.501 \rceil$. On the $4\times5\times5$ metric sheet the fix reduces bounded void-fill from 1,268,329 pixels to 3,707 while remaining bit-identical on integer-lattice tests. The episode is a useful illustration of why void-filling is dangerous as a default: the bug had produced a smooth, plausible, entirely false green smear over a million pixels of real geometry, and only the discrepancy between two representations exposed it.

Layers. Where the papyrus has delaminated, two plies legitimately occupy nearly the same (u, v), and both are real. Hard layer conflicts — overlap exceeding 5% of the smaller chart — are separated into distinct sheet bakes; soft boundary kisses remain in the primary sheet but are recorded edge by edge. The final first-cover composite prefers direct coverage from any layer over an earlier layer's generated fill, and writes a per-pixel label. On the fixed-topology $4 \times 5 \times 5$ result, secondary layers recover 389,628 pixels the primary alone misses and occlude 1.25 million pixels of genuine duplicated ply coverage; every face is retained in a layer manifest, so nothing is discarded, only ranked.

The output mask. Fitted pixels are kept unconditionally. A *filled* pixel is kept only when the final bake classifies it as papyrus and its Sander distortion is below 2 — its final measured state decides, not the fact that a PDE produced coordinates for it. Islands smaller than 64 pixels are removed as pepper, and the innermost region below radius 64 is excluded regardless of appearance, because pitch is not measurable there and the winding coordinate of Equation (1) is consequently undefined.

This policy yields 83.2% coverage, 3.02% dark, and no $\geq 4\times$ pixels on $4 \times 5 \times 5$. On $10 \times 10 \times 10$ it yields 70.1% coverage — at parity with the previous 70.6% supported sheet — while improving dark from 4.58% to 3.59%, Sander $p95$ from 2.73 to 1.45, and the $\geq 4\times$ tail from 2.53% to 0.03%. The full rectangle is retained beside the mask as a diagnostic. It is never the sole deliverable.

10 September 2026 Public Release

The September 30 release provides one generic configuration and a native sheet-assembly command. Parameter-search scripts, experiment and replay commands, external automation drivers, and the bottom-up hierarchical welding and repair runners have been removed. The versioned C API, including its existing option structures, is preserved. Direct grid meshing, registration, unrolling, and the optional quad-ribbon path remain available. The research results elsewhere in this report describe the August campaigns; the run in this section is the measured result of the public configuration.

Table 4. Public-configuration reference check, September 30, 2026. Area coverage is measured against the recorded source inventory, not the raster's bounding rectangle.

Measure	Result
Source faces accounted for	916,995 / 916,995
Represented source faces	646,199
Represented source area	47.385%
Placed / unplaced / excluded charts	356 / 406 / 21
Invalid faces / strict metric failures	0 / 0
Overlapping triangle pairs	171
Represented continuity components	233
Geometry qualified	No

10.1 One Configuration and a Staged Assembler

`configs/default.json` supplies the shared configuration, with `configs/axis.csv` carrying the reference axis. The initial calibration uses PHerc0139: axis $(y, x) = (3405, 2878)$, wrap spacing 9.5 coarse voxels, eight worker threads, and a 900-second repair budget. The input pile determines the region; the configuration does not impose a hidden crop. Users must set local CT and executable paths and replace the geometry calibration when changing datasets.

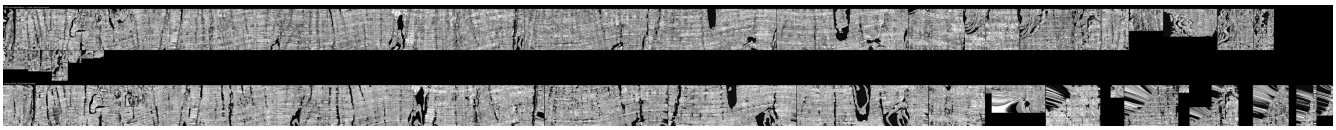
The command `sheet_assemble` accepts a mesh-pile directory, an output directory, and an optional configuration path. It cleans and flattens charts, measures cross-cube relations, resolves their poses, places them in a shared winding-aware domain, admits additional charts, and repairs UV seam, contact, and metric defects. An audit then accounts for the recorded source inventory and checks the resulting embedding before export and CT baking. Saved placement checkpoints support resumption; named stage stops and review commands support inspection. Repair changes the developed coordinates while preserving the represented source geometry in three dimensions.

10.2 PHerc0139 $4 \times 5 \times 5$ Reference Check

The check uses the saved `head_39e72d0_20260826` mesh pile: 100 cubes of 128^3 voxels spanning the half-open bounds $z = [4352, 4864)$, $y = [3072, 3712)$, and $x = [2560, 3200)$. Assembly completes through audit. All 916,995 source faces are accounted for, and the represented source vertices are unchanged. Table 4 reports the remaining limitations.

The configuration is a usable starting point, but this result is *not geometry-qualified*: its coverage is below the audit's 85% minimum and contacts remain. A successful command exit records processing completion, not geometric acceptance. This check does not reproduce the earlier checkpoint-58 atlas or the August quad-ribbon campaigns, which use different inputs and workflows.

Texture and evidence. Figure 10 is baked directly from the saved repaired VMESH without repacking its pieces. Sampling uses one pixel per coarse voxel, nine samples over ±4 voxels along the surface normal, and a fixed 8-bit display window of [47, 193]. This window is an explicit preview setting; it does not change the assembly audit. All 100 matching RAW cubes contain CT data. The baker leaves 18,456 boundary faces unpainted because their sampling range reaches



(a) Fixed-window RAW comparison: V83 (top) and the progressive baseline (bottom).

(b) The corresponding Sander distortion view; red tails concentrate in the baseline's folded continuation.

Fig. 9. Transactional solid-ribbon result. Against the same RAW source, the mature V46/V83 lineage lowers dark area, eliminates multiple coverage, and reduces the $\geq 4\times$ distortion population by roughly $320\times$ relative to the progressive baseline; V83 then supplies the measured convergence point of the residual loop.

absent neighbouring cubes; every painted face has the required RAW support. The native coverage mask separately labels 38,630 pixels of bounded raster crack or pinhole fill. The committed release evidence index retains the complete audit, bake coverage report, raster grid, input and artifact hashes, and commands.

Validation. The Windows Release CMake build, optional native quad-ribbon build, install check, and all 16 CTest checks pass. This validates the build and interface contracts; it does not convert the partial sheet into a qualified unwrapping.

11 Interfaces and Downstream Integration

In brief: how the pipeline is invoked, how it embeds in other software, and what a downstream tool is told about the geometry it receives.

Remote-volume workflow. The public `scrollunwrap` entry point streams an OME-Zarr region from anonymous or authenticated S3 storage, sends the selected voxels to the in-memory C mesher, and can continue through preview and `tifxyz` output without first materializing a TIFF cube grid. It also supports local arrays, mesh-only operation, independent cube and flattener timeouts, and graceful cube skipping. All outputs carry the true (x, y, z) origin and scale, so the resulting OBJ is immediately viewable and its atlas indexes the source volume rather than an internal cube convention.

Native sheet command. The public entry point `sheet_assemble <pile_dir> <out_dir> [config.json]` defaults to `configs/default.json`. Named stage stops, placement resumption, and checkpoint review are retained; experiment and replay cases are removed. The optional `quadribbon <mesh_dir> <out_dir> [config.json]` command reads the same configuration for the quad-ribbon workflow described above. The reference axis accompanies that file, and CT and executable paths remain explicit local inputs. Section 10 records the public configuration's measured result and qualification boundary.

Embeddable implementation. One versioned public C header exposes cleanup, split, topology, reconstruction, remeshing, weld, and full volume-to-mesh operations. A consumer can use a static library, a shared library, or a runtime-loaded ABI table; cross-runtime allocation, structured status, progress, timeout, and cancellation are all explicit. The CMake build is exercised both as an embedded consumer and on Linux/GCC. The MLS collapse can run through the reference C path, OpenMP, an optional FP32 CUDA backend, or a research CubeCL CUDA backend; every backend applies the same guards and is required to produce the same geometry, so a backend may be faster but never more permissive.

Application handoff. Clean local components enter `villa-patch`. ScrollFiesta preserves its intermediate ribbon, quilting, and snap artifacts, and Villa performs the authoritative final flatten. Larger uncertain surfaces enter VC3D through `ScrollFiesta: Generate Spiral Hints`, which creates a new `scroll-hint/unverified` segment, supports progress and cancellation, reloads the surface, and can add it to the active spiral fit; it never overwrites its source. The separation matters because the two consumers want different things: a patch tool wants geometry whose wrap identity is settled, and a whole-scroll fitter wants candidate surfaces whose identity is still open.

Output labels. Geometry and textures distinguish released prediction, direct fit, harmonic or cylindrical fill, moved-by-CT vertices, suspended order relations, primary and secondary ply, bounded raster void-fill, and masked output. A consumer receiving a partial result can therefore use it without confusing a generated continuation for observed papyrus, and a consumer that disagrees with one of our thresholds can re-derive its own conclusion from the labels rather than from the picture.

12 Results

The results in this section retain their August 2026 experimental scope. The September public-configuration audit and CT preview are reported separately in Section 10; the coverage measures and input lineages should not be combined.

In brief: we report the stages separately — sheet coverage, reading order, winding energy, metric distortion, collision state, and raster quality — because they fail in different ways, and a single aggregate would hide the one failure that matters. The experiments that did not work are reported alongside those that did.

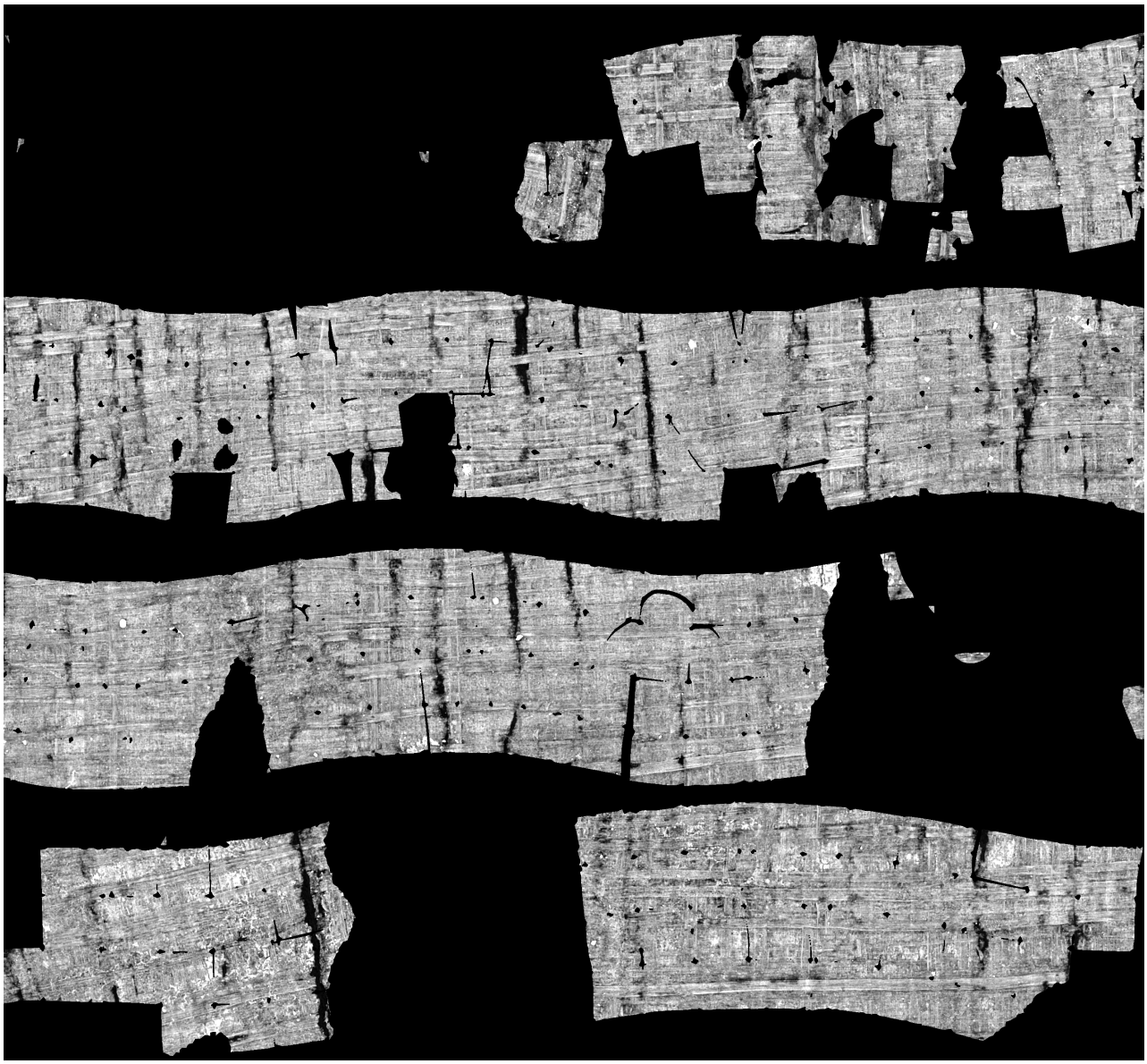


Fig. 10. **The public-configuration PHerc0139** $4 \times 5 \times 5$ **sheet.** CT sampled through the saved repaired VMESH, with the native $11{,}170 \times 642$ raster wrapped into four consecutive rows for viewing. The row breaks are display breaks, not inferred seams. The sheet is partial (47.385% source-area coverage), and 171 overlapping triangle pairs remain. Black includes unrepresented material and boundary faces whose sampling range reaches beyond the available CT cubes. This is the September reference check, distinct from the August exhibits.

A single aggregate score is the wrong instrument for this problem, for the reason Section 2.3 makes concrete: the decisive failure — a surface that has stepped onto the neighbouring turn — is invisible to every geometric quality measure, so any score computed from geometry can be improved by committing it. Each stage below is therefore measured against a quantity it could have failed, and where a stage is judged against CT, the fit being judged never saw that CT.

12.1 Sheet Coverage and Reading Order

The sheet books of Table 3 carry more than 70% of evidenced material on every rung. Count and material diverge sharply at full scale: the 21^3 book accepts 6,188 of 6,500 lifted sheets, yet its largest single sheet carries only 0.15% of the material and the 70% prefix requires 5,493 tracks. That fragmentation is a property of the evidence, and the held-out CT check catches both local bad interpolation and large prediction phantoms.

Winding assembly scales more favourably than the geometry count. The 21^3 main component orders 15,457 tracks over 112 winds at 5.0% residual conflict weight, assigns 99.3% of accepted material, and suspends the order claims of 174 tracks; the smaller $4\times5\times5$ and $10\times10\times10$ rungs need none. The interpretation is geometric: radial order survives fragmentation that geometry does not, because it is measured from separation across a local pitch rather than from connectivity. The remaining difficulty is longitudinal, not radial.

12.2 Probabilistic Winding Results

The generalized winding field supplies an independent, noisy absolute cue to the discrete relation graph. In a full-field probe of 313,615 vertices, the two-sided field exhibits the clean-sheet unit jump at 310,686 vertices (99.1%); the remaining 0.9% are discarded principally by local orientation cancellation. This measures local field behaviour, not an absolute turn label. Absolute gauge is still fixed once per relation island, and the field is downweighted or discarded where its within-island calibration is weak.

On a fresh $4\times5\times5$ PHerc0139 ball-pivoted pile (882,219 vertices, 1,711,155 faces, and 783 disk components), the standalone winding path contracts 783 patches into 25 continuation components and four relation islands. Of 2,620 measured relations, 1,698 enter the MRF. Two recentered graph-cut rounds change four component labels, reduce the measured energy from 221.369 to 200.486, and leave continuation and order satisfaction at 0.99971 and 0.99816, respectively; the complete winding-only run takes 10.8 seconds. Mean conditional label confidence is 0.99992, but we report it only as local sharpness with neighbouring MAP labels fixed, not as an exact global marginal.

The outer conflict-penalty experiments establish the present boundary. An unguarded 24-round run is rejected because its winding span expands from 14.13 to 32.41 turns — the penalties bought local agreement by inventing turns. A broad sweep finds 21,420 cells in which spatially remote geometry lands in the same developed position; winner- and context-lock variants reduce this to 11,099/10,214 only by breaking the relations the labels were built from, dropping continuation/order satisfaction to 0.95789/0.66605 and 0.95580/0.80329. Restricting attention to cells with at least four mutually remote claimants leaves two cells and four losing claims. The one-bin variant clears both and preserves winding span, but creates 42 residual relation conflicts and raises MRF energy from 200.486 to 1,119.330; requiring the loss in two cells makes no exclusion at all and preserves the base labels bit-for-bit. The base winding result stands; no penalized configuration improves duplicate claims without damaging the relations.

12.3 Metric Projection Across Frozen Rungs

Table 5 reports the common fixed-topology projection engine on three frozen inputs. It is explicitly *not* an end-to-end scale table: $4\times5\times5$ carries authoritative phase from the modern fit, $10\times10\times10$ uses an older solid input, and 4×21×21 uses a carry-registered ribbon. What the comparison does test is whether one metric projection preserves each input's established order without a new gauge search — that is, whether the projection is a function of its input rather than of its tuning.

The $4\times5\times5$ map has $|du/ds-1|$ at numerical precision and transfers every vertex and face. Phase placement moves 66 detached components from an atlas-like packed input into physical order, and secondary layers recover every marked missing region while keeping 1.25 million duplicated-ply pixels out of the primary view (Fig. 7). The $4\times21\times21$ carried frame is already close to ideal and remains so. The $10\times10\times10$ frame is substantially compressed and rough — the common metric map expands its mean rate from 0.777, and residual cross-row jitter and source-fused continuation remain — but even there it improves the retired solver's link $p95$ from 89.6 to 48.3 and alignment RMS from 290 to 150 *without changing carried order*, which is the property being tested.

The unified pre-weld path introduces the robust consistency system because its independent cube observations contain cracks the frozen projection inputs do not. At $4\times5\times5$, 2.04 million unknowns and 12.2 million nonzeros factor in 8–11 seconds; the crack rows reduce local slips from ±20–25 to ±1.3 pixels, and median texture seam shift falls to 3 pixels. The canonical 4×5×5 sequence completes end to end. A separate $21\times21\times21$ campaign reaches the maintained pre-fit concatenation boundary with 9,110 unique meshes and intentionally stops there; the fixed-topology comparison numbers above remain attached to the experiments that produced them.

A preliminary post-projection experiment asks whether continuous distortion can improve without reopening the discrete winding decision. On the boundary-complete 882,219-vertex, 1,711,155-face pile, eight UV-only symmetric-Dirichlet sweeps with a two-voxel displacement cap take 8.1 seconds. Area-weighted mean anisotropy falls from 1.761554 to 1.628350 and symmetric-Dirichlet energy from 63.28M to 56.95M; the independent primary-sheet raster improves condition-number median/p95/p99 from 1.408/2.661/6.014 to 1.320/2.413/5.278. Mean displacement is 0.190 voxels, no face changes its input orientation, and the 24,340 faces already over the stretch bound are left untouched. This is an order-preserving result on one rung, not a pipeline default or a scaling claim.

12.4 Attachment, Distortion, and Output Quality

Figure 9 and Section 9.3 give the mature $4\times5\times5$ attachment result: V83 carries 8.168 million faces with zero coverage fault, 3.162% dark pixels, Sander $p95/p99$ 1.190/1.380, and 428 samples at $\geq 4\times$. Against the fixed progressive baseline, the V46/V83 lineage removes multiple coverage entirely and has roughly 320 times fewer $\geq 4\times$ samples. The loop terminates by rejecting its own final proposal, every measured quality tail having worsened, which locates the fixed point by measurement rather than by a visually chosen iteration count.

Provenance. One piece of bookkeeping earns its place among the results, because it changes what a coverage number means. Every vertex and pixel carries a label separating directly supported geometry from harmonic or cylindrical interpolation, additive prediction bridges, and vertices moved by CT attachment. The classes behave differently, and averaging them hides the difference. On $10\times10\times10$, restricting the output to fitted material plus interpolation that survives its final CT and distortion state gives 70.1% coverage at 3.59% dark pixels, Sander $p95$ 1.45, and a 0.03% $\geq 4\times$ tail; the full rectangle — which contains continuation across wraps the source prediction

Table 5. Fixed-topology metric projection on three frozen upstream states. Fill is reported only where the source run defines a comparable raster; $4 \times 5 \times 5$ instead reports the composited painted population. These are three different inputs through one engine, not one pipeline at three scales.

Grid	Vertices	Faces	Output coverage	Length RMS	Link $\lvert\Delta U\rvert$ med./p95	Time
$4 \times 5 \times 5$	2.040M	4.033M	6.82M painted px	0.00452	0.300 / 1.15	9.6 s
$10 \times 10 \times 10$	1.512M	3.012M	60.8% fill	0.0216	6.02 / 48.3	6.8 s
$4 \times 21 \times 21$	6.731M	12.946M	68.1% fill	0.00415	0.205 / 1.27	30.8 s

had already fused — reports a near-identical 70.6% coverage at 4.58% dark, Sander $p95$ 2.73, and a 2.53% tail. Coverage is at parity; every quality measure is not, and only the labels distinguish the two. On $4 \times 5 \times 5$ the same restriction yields 83.2% coverage, 3.02% dark, and no $\geq 4\times$ samples.

12.5 Measured Dead Ends and Remaining Failure Modes

Five negative results narrowed the method, and each one fixed a design decision described earlier in the paper. What follows is in each case an account of what the objective actually optimizes, because in every one of these the implementation was correct and the objective was answering a different question from the one intended.

- **Winding-conflict penalties (Section 8.1).** A near-hard unary forbidding label k for component c has a cheapest escape, and it is the wrong one. The pairwise factors are local, so translating an entire weakly connected component several turns away in k costs almost nothing, while satisfying the forbidden assignment in place costs the relations that hold it. The penalty's least-energy solution is therefore to invent turns — which is exactly the failure it was added to prevent, and precisely what the unguarded run does when the span goes from 14.13 to 32.41. Restricting which cells may issue a penalty bounds the damage without changing its direction: broad winner and context locks halve the conflicting cells by breaking the relations the labels rest on, the one-bin variant clears its two four-claimant cells at more than fivefold the energy, and the two-bin variant issues no penalty at all. The lesson is about the relative strength of two measurements, not about tuning: a coincidence in developed position is weaker evidence than a measured radial relation, and an objective that lets the first overrule the second will spend the second to satisfy it.
- **The geometric slab splitter.** It carves 11.4 million prediction pixels (4.1% of foreground) along visually convincing medial ridges in 2.8 seconds and separates 0 of 246 measured fused runs. The reason is a mismatch of scales between two operators: a cut in the probability volume is one voxel wide, while reconstruction spans gaps up to the pivot radius $\rho = 1.2$ and the MLS collapse first pulls points from both sides of the cut toward a common surface. A separation the mesher cannot see is not a separation. Cutting in the voxel domain can only work if the cut exceeds the reconstruction's own reach, and a medial ridge never does.
- **The tabu winding atlas (Appendix C.8).** It reduces 373,536 exact UV overlaps to 1,874 — a 99.5% reduction — while tearing 41 usable weld relations. It remains a reproducible baseline, and its failure is the reason the current system does not let overlap minimization choose winding identity: overlap count is a geometric objective, and by Section 2.3 a geometric objective cannot see the distinction it was being asked to decide. At $10 \times 10 \times 10$ the search is net-negative under every configuration tested, and on both tested rungs the no-search control plus u-packing reaches higher coverage than any search configuration (71.7% and 70.1%): the deficit the search was introduced to close was packing, not placement.
- **The full elastic shell (Section 9.2).** It reduces 147,728 residual contacts to 4,073 after snap and then stops. Each contact contributes a unilateral clearance constraint on a scalar radial field, and on a tightly wound coil the constraints are not independent: opening one penetration presses the sheet against the next turn, whose constraint then forbids the motion. Past a certain contact density the feasible set is small enough that no admissible displacement remains, so the residue is a property of the constraint system rather than of the number of rounds. Contact pruning and the newer pre-weld input are the next controlled variables.
- **Generated continuation at $10 \times 10 \times 10$.** The fitted region is high quality, but continuation crosses wraps that were already fused in the source prediction. A parameterizer receives a surface, not a history: where two turns arrived already joined, there is no residual in any energy it minimizes that distinguishes that join from a real one, and a smooth continuation across it is the correct answer to the question it was asked. This is the direct motivation for the division of labour with the companion pipeline: the distinction is representable in the volume, where the two sheets still have different radii, and is gone once they are one component. No amount of care downstream recovers it, which is why the prediction volume is completed before it reaches us rather than repaired after meshing.

One pattern runs through all six. In each case an operator was asked for a decision whose evidence had already been destroyed by an earlier representation change — reconstruction closing a one-voxel cut, an overlap count standing in for a measured relation, a parameterization receiving two turns already fused. The gains came from moving each decision to the domain where the quantity it turns on is still present: winding phase in the mesh, radial order in the relation graph, arc length in the ribbon. Strengthening the downstream optimizer never helped, because the information it needed was not in its input.

13 Conclusions

The September release makes the native sheet workflow available through one generic configuration while retaining the public C API. The reference $4 \times 5 \times 5$ check produces the partial CT-textured sheet in Figure 10: 47.4% source-area coverage, 171 overlapping triangle pairs, and unchanged represented source geometry. It remains unqualified. The research conclusions below concern the distinct August campaigns and do not imply that the public defaults reproduce them.

The recurring difficulty in virtual unwrapping is that the decisive error — a surface that has walked onto the neighbouring turn — is invisible to every measure a mesh can supply. ScrollFiesta's response is to carry the quantity that *can* see it. A branch-cut-free winding coordinate, constant along one wrap and incremented by one at each step outward, is computed alongside the geometry and used as the decision variable at every scale: as a radial bound on a voxel-domain bridge, as a growth guard on a pivoting front, as the integer relation between two sheet tracks, as the label set of a Markov random field, and finally as the horizontal axis of the flattened page.

The three contributions of this round follow from that. First, identity is represented apart from geometry: sheet tracks carry pairwise integer winding relations, composed through a maximum-support forest with integer subtree-shift loop closure, so a remeshing or welding decision cannot quietly change which turn a piece of papyrus belongs to. Second, the metric ribbon *consumes* that turn assignment rather than searching for one, so a low-distortion or low-overlap solution cannot buy itself a different physical identity. Third, winding is inferred instead of rounded: a two-sided generalized winding number field supplies robust per-component unaries, local continuation and radial-order observations supply pairwise factors, and alpha–beta swap graph cuts solve the resulting integer MRF.

The representation scales. Sheet books carry more than 70% of evidenced material at every rung through $21 \times 21 \times 21$, and the largest winding assembly orders 15,457 tracks across 112 winds while retaining 99.3% of assigned material. The metric lane then shows that a solid, readable quad ribbon can be obtained without letting an atlas search own physical identity: exact arclength against one shared metric-rate map establishes placement, Huber-IRLS sparse least squares closes measured cracks and contacts, an exact contact response untangles the coil, and CT attachment moves vertices onto the papyrus band while keeping interpolated material labelled apart from measured material. On the strongest $4 \times 5 \times 5$ campaigns, all 4.033 million fixed-topology faces are conserved, every marked missing region is recovered from a supported layer, and the attachment loop reaches a zero-coverage-fault fixed point at Sander $p95/p99$ 1.190/1.380.

The probabilistic winding result resolves the long-standing ambiguity more directly than a smooth atlas can, because it puts the integer where the ambiguity actually lives. On the fresh $4 \times 5 \times 5$ pile, the clean-sheet field jump holds at 99.1% of queried vertices; the MRF changes four component labels, lowers its energy from 221.369 to 200.486, and retains 0.99971/0.99816 continuation/order satisfaction. The failed penalty experiments are equally informative: broad variants halve the duplicate-claim cells only by damaging the relations that produced the labels, and the targeted one-bin rule clears its two four-claimant cells at more than fivefold the energy. A count of spatial collisions is a weaker signal than a measured relation, and the experiments say so numerically.

The negative results are part of the contribution rather than an appendix to it. No conflict-penalty variant improves both of its objectives; a slab splitter separates 0 of 246 fused runs despite a convincing visualization; a tabu atlas removes 99.5% of UV overlaps while tearing 41 weld relations; and generated $10 \times 10 \times 10$ continuation crosses wraps its source had already fused. Read together they locate the remaining failure precisely: missing material and large-scale wrap identity must be settled before geometry is filled or flattened, because no downstream optimizer can recover an identity that is absent from its input.

Limitations. This is not a complete automatic unwrapping of a Herculaneum scroll. The August one-command quad-ribbon campaign completed its canonical $4 \times 5 \times 5$ run from the pre-weld pile, but the larger metric results come from frozen inputs of earlier lineages and are not an end-to-end scaling curve. The 21^3 sheet book supplies radial order but no comparable measurement for longitudinal continuation across a large hole — which is why Fig. 2 draws that link as absent. The elastic shell retains 4,073 post-snap contacts on the full sheet. The crushed inner core has no measurable pitch, so the winding coordinate is undefined there and the region is excluded regardless of how it renders. The targeted conflict penalties have not resolved their two cells without sacrificing stronger continuation and order evidence.

Next experiments. The remaining work follows directly from those limits: obtain an independent measurement of absolute turn error, which the MRF currently cannot supply for itself; redesign the conflict penalties so they reduce remote duplicate claims without expanding span or lowering relation satisfaction; carry the global fit forward from the $21 \times 21 \times 21$ pre-weld pile; and feed same-wrap continuation relations into the ribbon fit without merging components — the one link Fig. 2 still draws as absent.

ScrollFiesta is not yet a read scroll. What it offers is a formulation in which the question that decides the answer — which turn of the roll a piece of papyrus belongs to — is represented explicitly, measured where it is still observable, and never delegated to an optimizer that cannot see it.

A Developments Since the July 31 Update

This is the only part of the paper organized by date. The August research account describes the methods and experiments from that period; Section 10 separately records the September public release. This appendix records what changed, module by module, between August 1 and August 31, 2026, relative to public commit `4f43cfc`. It is intended for a reader who has the July 31 paper and wants the delta rather than the whole story. A complete pre-edit copy of the July paper is retained beside the working repository, and the retired methods referenced below remain documented in Appendix C. Prediction-domain work from the same period — cross-resolution distillation, connection mining, and additive bridge filling — has moved to the companion *Socratic Method* repository and is recorded there instead.

- **Slice tracing and in-plane fusion cuts.** Z-plane skeleton paths are associated across axial shards into material tracks. A new along-curve ownership test splits paths that travel from one wrap through a prediction fusion onto another and then reruns bootstrap association. Coverage remains unchanged while fused paths become separately checkable pieces.
- **Sheet books.** Batch lifting freezes geometry before held-out RAW, records direct and interpolation masks, and rejects crashes, black phantoms, normal flips, and unsupported interpolation. The books pass 49/50 sheets at 83.7% material on $4 \times 5 \times 5$, 385/400 at 74.3% on $10 \times 10 \times 10$, 1,104/1,150 at 71.9% on $4 \times 21 \times 21$, and 6,188/6,500 at 71.4% on $21 \times 21 \times 21$.
- **Winding assembly.** Local signed radial contacts snap to integer relations only within 0.25 turns of a pair-specific pitch. A maximum-support forest and strict subtree-shift repair establish reading order without fusing geometry. The largest rung orders 15,457 tracks across 112 winds, retains 99.3% of assigned material, and suspends the order of 174 contradictory tracks.
- **Probabilistic winding field and MRF.** A two-sided generalized winding-number field measures a mean and clean-sheet jump at mesh vertices, robust component summaries become calibrated unary terms, and continuation/order observations become pairwise factors in an integer-label MRF. On the fresh $4 \times 5 \times 5$ pile, 99.1% of 313,615 probed vertices exhibit the unit-jump law; two graph-cut rounds change four labels and reduce energy from 221.369 to 200.486 while retaining 0.99971/0.99816 continuation/order satisfaction. A standalone winding-only command writes geometry-identical output and field, jump, label, confidence, and JSON sidecars.
- **Guarded winding-conflict penalties.** Sparse near-hard unaries can forbid an exact component/label assignment when spatially remote geometry repeatedly claims the same developed cell. Broad winner/context locks reduce 21,420 conflicting cells to 11,099/10,214 but damage relation satisfaction. Restricting to cells with at least four claimant clusters finds two. A one-bin trial clears them using 20 exclusions but creates 42 residual relation conflicts and raises MRF energy to 1,119.330; requiring two cells makes no exclusion and preserves the base labels. Synthetic tests pass; no real-data configuration resolves the conflict without spending more relation evidence than it recovers.
- **Sheet compositor.** Continuously lifted phase replaces principal-angle folding; exact metric offsets join accepted same-wrap continuations. The output is a native-resolution texture plus contributor and label maps, with a separate page-layout review image. Unpainted graph bridges may transmit order but never contribute texture.
- **Solid quad ribbons.** A dense C implementation builds a regular quad lattice inside each material component, fills interior holes by harmonic or cylindrical continuation, and never fills between components. Direct, interpolation, filled, and seam vertices remain separately labelled. VMESH, OBJ, topology, snap transaction, and streamable snip outputs make the large meshes inspectable.
- **Recto attachment.** Snap now alternates winding-aware CT targets, robust vector quilting, smooth displacement, correspondence remeshing, fixed-window RAW bake, and per-component selection. Exact collision, orientation, coverage, darkness, and Sander measurements accept or reject each proposal. V83 is the measured $4 \times 5 \times 5$ fixed point, not a hand-selected iteration.
- **Phase-ordered elastic shell.** Exact triangle contacts create a persistent phase and radial-order set; a screened radial field opens penetrations, and settle-to-rest returns the sheet toward measured geometry without pass-through. Parallel exact tests, active bands, frontier dilation, confidence-preserving coarse grids, and relative stopping reduce full-sheet wall time from a 16.5-hour non-converged predecessor to 7 minutes 29 seconds.
- **Final-state quality masks.** The output now retains fitted pixels and only those filled pixels whose final RAW and distortion states support them, removes small pepper islands, and excludes the pitch-ambiguous core. At coverage parity this lowers $10 \times 10 \times 10$ dark rate from 4.58% to 3.59%, Sander p95 from 2.73 to 1.45, and the $\geq 4\times$ tail from 2.53% to 0.03%.
- **Solver-free metric projection.** A controlled fixed-topology path replaces floating chart gauges with exact chain arclength, one shared metric-rate map, robust median offsets, and rigid detached-island placement by authoritative phase. The $4 \times 5 \times 5$ run conserves all 4,033,291 faces, recovers every marked layered dropout, and completes parameterization in 9.6 seconds without Krylov iteration.
- **Guarded post-projection Sander pass.** A preliminary UV-only experiment on the boundary-complete $4 \times 5 \times 5$ pile runs eight symmetric-Dirichlet sweeps under immutable input-orientation, per-face stretch, and two-voxel displacement guards. In 8.1 seconds it lowers mean anisotropy from 1.761554 to 1.628350 and raster condition-number p95 from 2.661 to 2.413 without changing any face's input orientation. The 24,340 faces already over the stretch bound are left untouched; the pass is not yet a default or a scaling result.
- **TAUCS IRLS consistency.** The active pre-weld fit now assembles metric, vertical, contact, and crack rows into normal equations, solves by supernodal multifrontal Cholesky, and applies Huber IRLS to exact residuals. A 2.04-million-variable, 12.2-million-nonzero system factors in 8–11 seconds; 28,597 crack ties reduce ±20–25-pixel slips to approximately ±1.3. The matrix-free CG and its ad-hoc compensation passes were removed from this lane.
- **Raster and layer correctness.** A half-texel face-bound bug at generic sub-voxel UV phases was fixed, reducing bounded void-fill on the metric $4 \times 5 \times 5$ sheet from 1,268,329 to 3,707 pixels. Hard ply conflicts become separate layer sheets; soft boundary kisses remain recorded; direct coverage from any layer outranks earlier generated fill in the final composite.
- **One quadribbon command.** `quadribbon` takes a mesh directory and an output directory, replacing the experimental driver stack and enforcing one order: concatenate the pre-weld per-cube pile, fit and optimize metric coordinates, untangle, re-optimize, bake layer sheets, and composite. It resumes by stage, uses one production configuration, and retains stage-local tools only for diagnosis. The canonical $4\times5\times5$ full run now completes end-to-end. The $21\times21\times21$ run has completed the maintained pre-fit boundary through stage-1 concat; global fit remains unrun.
- **Measured retirements.** The current lane no longer uses global geometric welding as a mandatory precondition, the July tabu atlas as winding authority, the matrix-free consistency solver, or the geometric slab splitter. Their recorded results remain documented, and Section 12.5 gives the measurement that retired each method. The September release does not ship the experiment drivers.

B August 2026 Progress Exhibit Gallery

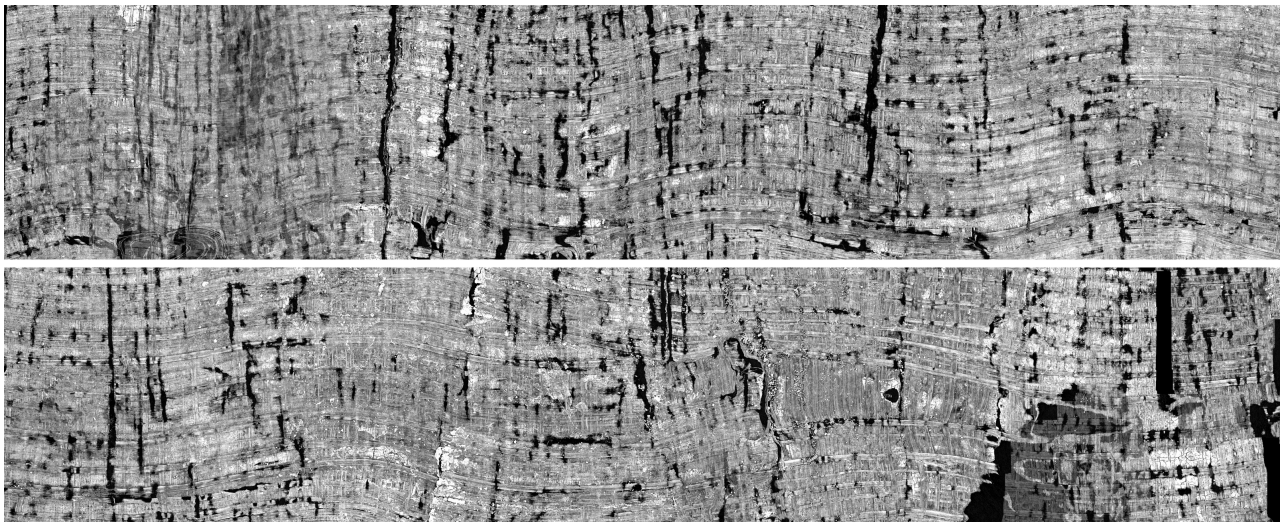
These exhibits retain their August 2026 scope. The September public-release sheet and its audit are shown in Section 10. References to current runs within this gallery refer to the August snapshot.

This gallery separates three meanings of scale that should not be collapsed into one claim. The $10 \times 10 \times 10$ hybrid ribbon is the largest coherent readable strip currently available; the $21 \times 21 \times 21$ sheet-book view is the largest reading-frame exhibit; and the new maintained $21 \times 21 \times 21$ geometry run is the largest canonical pre-fit pile. That last run stops deliberately after quadribbon concatenation and is not called a global fit.

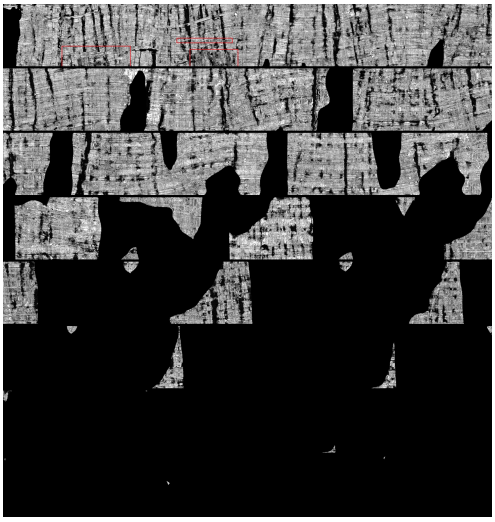
Exhibits. Fit state is kept separate from geometric scale so a large pre-fit pile cannot be mistaken for a solved global ribbon.

- **PHerc0139,** $4 \times 5 \times 5$**:** 100 cubes in the current integrated fitted and baked run; visual quality-mask faults fall from 41,407 to 34,938 (16%), with zero primary-sheet hard conflicts.
- **PHerc0139,** $10 \times 10 \times 10$**:** 16.444M vertices and 32.608M faces in the hybrid champion lineage; this is the largest coherent readable exhibit, not proof of the current general pre-weld fitter.
- **PHerc0139,** $21 \times 21 \times 21$**:** 9,110 unique world-frame meshes, 195.971M vertices, and 381.764M faces; the maintained 6.46-GiB stage-1 concat is complete with zero final failures, but fit was not run.
- **PHerc1203,** $5 \times 5 \times 5$**:** 3.984M vertices and 7.753M faces; 125/125 cubes complete with zero reject/fail, and 125/125 canonical meshes are 2-manifold with zero same-direction defects.
- **PHerc1447,** $5 \times 5 \times 5$**:** 2.170M vertices and 4.191M faces under the same guards; 125/125 cubes complete with zero reject/fail, and all 125 canonical meshes pass the same topology checks.

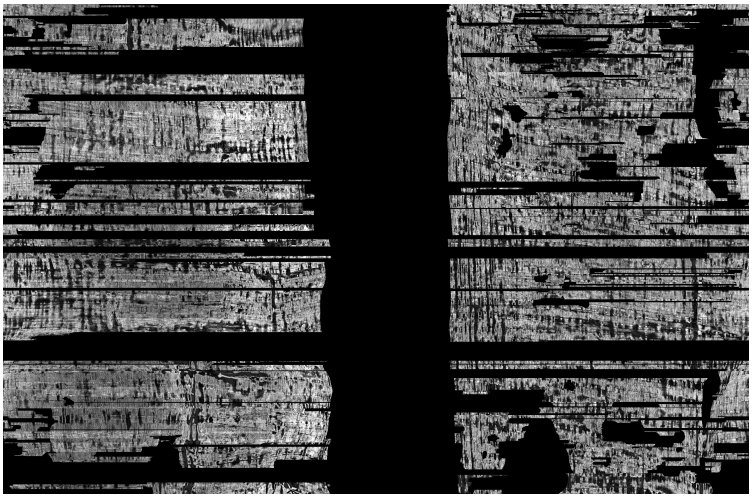
Reproducibility boundary. The current 21^3 concat log records an explicit stop after stage 1; its 9,110-row inventory balances exactly, and 12/12 overlapping cube meshes are byte-identical across the baseline and current runs. The grid, quadribbon, and manifold self-tests pass, and both prize piles were meshed with the winding guards armed. The multi-gigabyte VMesh is represented here by its checksum rather than copied into the paper repository.



(a) Largest coherent readable strip: the 10^3 hybrid champion.

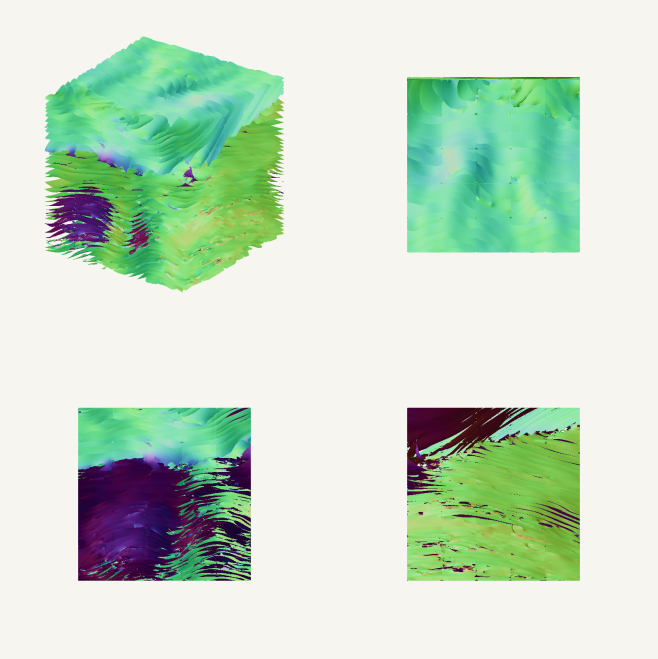


(b) Current $4 \times 5 \times 5$ integrated quadribbon page.

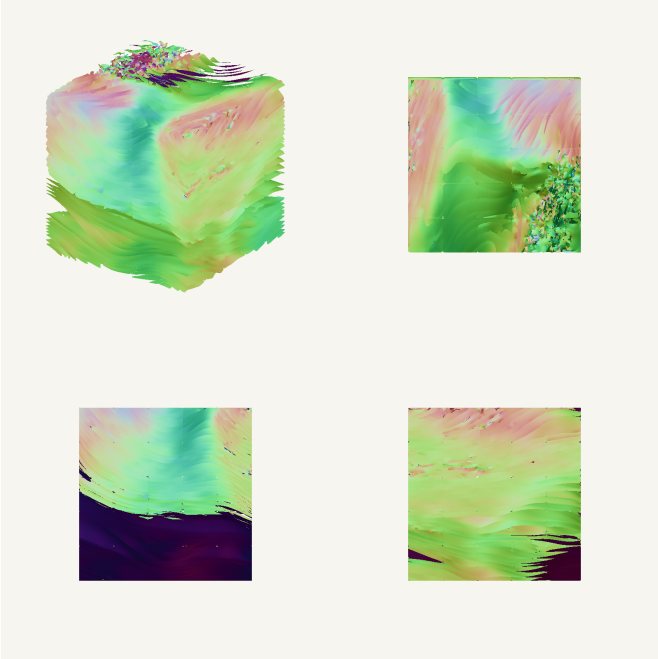


(c) One winding page from the full 21^3 sheet book.

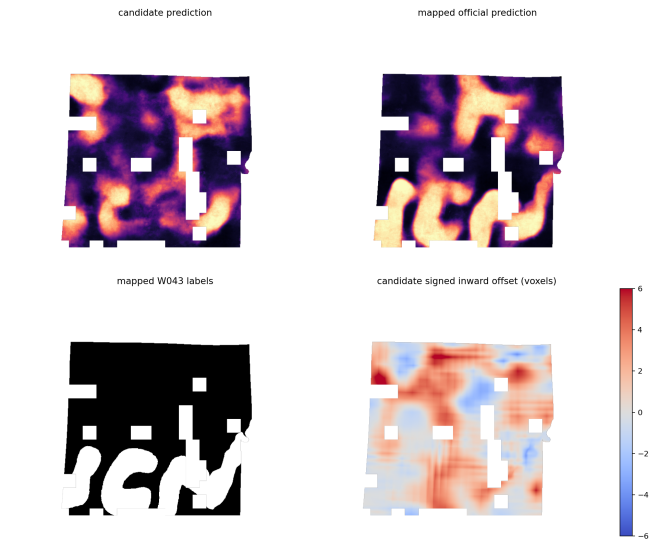
Fig. 11. PHerc0139 scale exhibit. The top strip is the visually strongest coherent result, while the lower-right panel reaches the largest evidence extent without pretending that separated tracks are one fitted surface. Black regions are places with no accepted track. Separately, the current 21^3 pre-fit mesh pile contains 9,110 unique world-frame contributors and stops before global quadribbon fit.



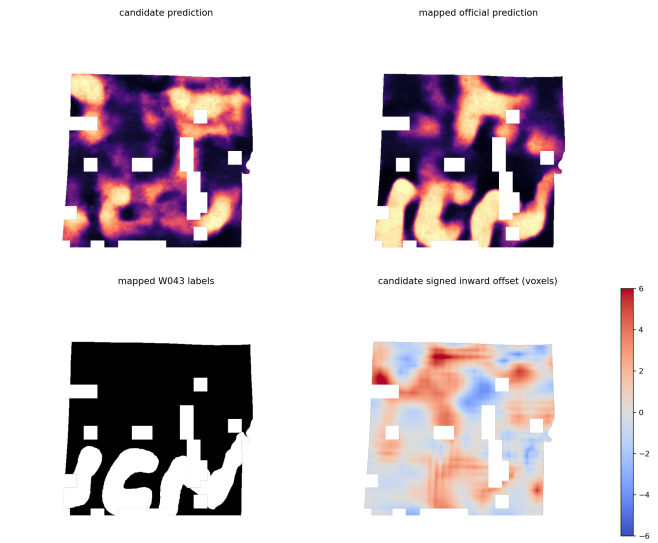
(a) PHerc1203: 125/125 current-pipeline cube meshes.



(b) PHerc1447: 125/125 current-pipeline cube meshes.



(c) W043 detector output on unchanged source geometry.



(d) The same detector after native 9.362-μm normal-only fitting.

Fig. 12. Cross-scroll transfer and downstream utility. The upper row shows the complete geometry-only 5^3 prize-target piles in isometric, top, front, and side views; colour is view-dependent normal shading, not a confidence or material label. No weld or global fit has been applied. In the lower row, native W043 fitting improves median signed offset from +0.810 to +0.395 voxels, normal-offset RMSE from 2.027 to 1.806, and supervised AUC from 0.6901 to 0.7098. This is a measured, modest improvement rather than oracle recovery.

The public exhibit preserves the corresponding counts, topology audits, stage boundary, and SHA-256 checksums in `submission_update/gallery/`. The 6.46-GiB PHerc0139 stage-1 VMesh is not duplicated into the paper repository; its checksum and cube-ledger totals are published instead. The prize regions contain prediction-derived geometry only, so no prize-scroll texture or ink claim is made.

C Parameters and Retired Baselines

These parameters describe the August research methods and retired baselines. Their experiment drivers are not part of the September public release. The shipped configuration is `configs/default.json`, with its measured reference check in Section 10.

This appendix records the exact numerical parameters referenced in the body. Sections C.1–C.4 give the active sheet-checking, winding, metric, fill, collision, and output-masking constants; Sections C.5–C.6 give the extraction and remeshing constants; and Sections C.7–C.10 document the recorded methods and parameters of retired baselines, and which Section 12.5 evaluates. Lengths are in coarse voxels and angles in continuously lifted turns unless stated otherwise.

Wherever possible a constant here is derived from a measured property of the object rather than tuned. The ∼7-voxel inter-wrap clearance and the ∼9.5-voxel pitch of PHerc0139 bound the ball radii, weld radii, collapse lengths, and neighbour distances below; the point spacing after MLS projection bounds them from the other side. Where a value was instead fixed by an experiment, the experiment is named alongside it.

C.1 Sheet Checking and Winding Assembly

A lifted sheet carries disjoint masks for direct prediction support, interpolation, and held-out evaluation. RAW is sampled only after fitted geometry and masks are frozen. Undefined or non-separating held-out correlation rejects the sheet, as do unmapped cells, out-of-support samples, normal-flip excess, and failed geometry checks. Book acceptance is material weighted: passing sheets must carry at least 70% of all evidenced cells on a production rung.

For an observed track pair (a, b), let Δr_{ab} be the median signed displacement along the local umbilicus ray and p_{ab} the robust median nearest-foreign spacing of their neighbourhood, clamped to $[0.8, 1.5]$ times the campaign pitch. The candidate relation is $q_{ab} = \mathrm{round}(\Delta r_{ab}/p_{ab})$ and is admitted only when its residual is at most 0.25 turns. Edges are sorted by sample support into a maximum-support forest. A subtree shift $k \in \mathbb{Z}$ is accepted only when it strictly reduces the total weight of disagreeing edges crossing that forest cut. Order suspension is capped at 5% of a component but has no positive floor; it removes only an order relation, never sheet geometry.

C.2 Probabilistic Winding MRF

The winding field samples w_+ and w_- at $v \pm \epsilon n$, with $\epsilon = 0.1$ local pitch clamped to $[0.25, 1.5]$. The mean is $(w_+ + w_-)/2$ and the oriented jump is $w_- - w_+$; clean sheet support has jump one. A sample receives reliability $\exp[-\frac{1}{2}((\mathrm{jump} - 1)/0.35)^2]$ and is discarded below 0.05. At most two million vertex samples enter the field summary. Each component's proposed correction is the reliability-weighted median of $-\mathrm{mean} - q$, with scale $\max(1, 1.4826\,\mathrm{MAD})$; the one-turn floor prevents a numerically sharp but globally compressed field from overpowering relations.

Field corrections are calibrated separately inside each relation island. Calibration requires at least three supported components, slope in $(0.125, 8]$, and $R^2 \geq 0.25$; its unary weight is capped at 0.25. Continuation factors receive four times their measured support after removal of the historical forest-priority offset, while order factors retain measured support; both have floor 0.25. The MRF uses Huber unary transition 2.5, fixed-point cost scale 256, residual labels $[-2, 2]$, six alpha–beta swap iterations per solve, and at most four recentered solves. A relation-island root fixes the gauge. Confidence temperature is 1, and local conditional sharpness below 0.35 is reported as unresolved.

Conflict exclusion bins claims within one relation island by 2-voxel axial position, 256-bin wrapped phase, and absolute turn. Samples within 0.55 local pitch are one claimant; different claimants must be separated by at least eight pitches and 48 voxels. A candidate cell must contain at least four claimant clusters. Ordinary two- and three-layer occupancy remains a quadribbon responsibility. A component must lose in at least one candidate cell and win in none before its exact current label receives a sparse unary penalty of 65,536. All non-losing context is fixed during that solve; moves remain within eight turns of the base state. The outer loop is capped at 16 rounds and retains only a strict reduction in candidate bins. The current one-bin configuration clears both real candidate cells at the cost of 42 residual relation conflicts; a two-bin variant makes no exclusion. This layer is experimental and not part of the production path.

C.3 Metric Projection and TAUCS IRLS

Fixed-topology projection computes exact XYZ arclength per row chain. A chain gauge is the median of $u_{\mathrm{seed}} - s$ over locally metric samples satisfying $|du/ds - 1| \leq 0.15$. One globally shared rate map F is built from robust cross-row rates in 32-voxel bins and reciprocal-integrated. Detached components may be placed only from the fit's continuously lifted phase sidecar through the dominant component's monotone phase-to-u map; sidecar-less inputs retain their carried placement.

The active consistency solve assembles rows $(x_a - x_b) \simeq d$. Metric rows use $d = ds$ and base weight $1/ds$. Vertical rows and face contacts use $d = 0$ with aggregate weights 32/length and 4/length, respectively. Crack ties probe the adjacent slice and are admitted only where both facing sides lack a face tie. One unit-weight pin fixes each connected null space. The lower triangle of $A^T W A$ is solved by TAUCS supernodal multifrontal Cholesky; at most three IRLS rounds multiply a robust row by $\min(1, c/|r|)$ from its exact residual, with Huber scale 1 for vertical drift and 6 for physical contacts. Island placement runs again against the solved field before a global re-zero.

C.4 Component-Bounded Fill, Collision, and Output Masking

Solidification allocates a quad lattice only inside one component's union of occupied row and column spans. Unknown world coordinates solve a four-neighbour harmonic Dirichlet problem, or an equivalent cylindrical radius/phase continuation, using that component's rim. A neighbour with another component label is not in the stencil; there is no inter-component fill state.

The collision shell uses exact triangle contacts to establish branch phase and radial order. Trials solve a smooth scalar radial response in an active band, reject new folds or orientation reversals, and may temporarily increase pair count only while phase-order violations decrease. The settle stage minimizes displacement from the pre-escape rest coil subject to the accumulated unilateral contacts. A research run may report the lexicographically best exactly-tested partial state; `-require-collision-free` fails closed unless every long-range stab reaches zero.

CT attachment proceeds per component, each move accepted or rejected on its own measurements. Each candidate retains a zero-motion option and is compared with its immediate input using complete RAW coverage, fixed-window dark count, Sander tails, UV and 3-D skips, multiple coverage, enclosed holes, exact collision, and orientation. A rejected component is bit-identical to its input. The output mask keeps all fitted pixels; a generated pixel additionally requires final class ok and Sander $D < 2$. Connected mask islands smaller than 64 pixels and all material at radius below 64 are withheld.

C.5 Per-Cube Extraction Constants

MLS projection. Each point v takes the neighbours of its point cloud within a Wendland-kernel radius of 12 (approximately twice the local sheet thickness, so the far face of a thick prediction still carries weight and collapses), estimates a normal n and weighted centroid c from them, and moves only along the normal, $v' = v - ((v - c) \cdot n)\, n$. Restricting motion to the through-thickness direction is what prevents the in-plane contraction of centroid replacement at high iteration counts. Five iterations are used at extraction and twenty when a separated piece is re-projected from its parent's original points; points within 0.25 of one another are then welded, and a point with fewer than four neighbours is left in place. Determinism across cubes requires only that the read halo exceed the kernel radius by one voxel (13), so every owned point sees its full two-sided support.

Ball pivoting and guards. The pivot radius is $\rho = 1.2$, about 1.55× the ∼0.77 projected point spacing: below ∼1.1× spacing, sparse sheets fall through the ball; above ∼2×, the empty-ball test crowds with cocircular intruders. Triangle quality is insensitive to ρ in this band (it is fixed by the point arrangement), so ρ acts purely as a coverage/anti-merge knob. The wall guard rejects a candidate whose face normal is tilted more than ∼70° from the mean vertex normal ($|\cos| < 0.34$) *and* whose spanning edge exceeds 2.0 — the conjunction fires on through-thickness fusion bridges and spares short-edged folds and rims. The per-seed growth guard integrates the branch-cut-free winding phase $w = r/p - \theta/2\pi$ along the advancing front and refuses a pivot once the accumulated phase leaves its tolerance band; the surface beyond is re-seeded as a new component. Non-separating handle loops shorter than 60 are severed after hole filling; a per-cube sheet has no legitimate scroll-scale handle, and the length bound is the safety margin.

C.6 CVT/RVD Remeshing Constants

The remesher minimizes the CWF objective [Xu et al. 2024], a centroidal Voronoi energy plus a normal-anisotropy term, $E = \lambda_{\mathrm{CVT}} \sum_i \int_{\Omega_i} \|x - x_i\|^2 \, dA + \lambda_{\mathrm{NA}} E_{\mathrm{NA}}$, where Ω_i is site i's restricted Voronoi cell on the surface and E_{NA} penalizes cell mass displaced off the local tangent plane, pulling samples onto creases. Each iteration solves, per site with fixed cells, the 3×3 system $(\lambda_{\mathrm{NA}} N_i + \lambda_{\mathrm{CVT}} m_i I)\, x_i = \lambda_{\mathrm{NA}} b_i + \lambda_{\mathrm{CVT}} m_i c_i$ (m_i, c_i the cell's mass and centroid; N_i, b_i the anisotropy normal equations), then projects x_i back to the source surface. Both weights start at 1 and λ_{CVT} decays by 0.95 per iteration to a floor of 10^{-3}; the pipeline runs 12 iterations, at which the decayed energy has converged. Uniform density places 0.0025× the input face count in sites (minimum 50). At weld time the seam band is refined to 3.5-voxel triangles — pinned by bridge priming, since a hinge ball reconstructs only when the front triangle's circumradius is below the bridge radius — and recoarsened afterwards by guarded collapse of band edges shorter than 5, a cap deliberately below the 7-voxel clearance so no collapse can fuse wraps. The pinned-boundary variant makes every patch boundary vertex an immutable site (bit-exact, so the patch stitches back conformally), seeds the interior at the target spacing while keeping 0.85× the local boundary segment length clear of the rim, and accepts a patch only if every pin is referenced, every consecutive boundary pair is still an edge, and manifoldness, component count, and Euler characteristic are unchanged; rejection is per ∼cube-scale tile, so one core fold cannot reject a whole seam.

C.7 Retired: Slice-Stroke Metric Solve

The unknowns are one developed coordinate u per slice sample plus one latent intercept per cross-section. Three row types feed the least-squares system: *arc-length rows* asking consecutive samples of a stroke to advance by their measured 3D arc length; *cross-section rows* tying each member sample to its cross-section's latent isovalue, weighted by a membership likelihood; and *continuation rows* joining two stroke endpoints on one slice with target $u_a - u_b = \tau \cdot (p_a - p_b)$, the tangent-projected physical gap, emitted only for mutually selected, sub-half-wrap joins. The radius-anchored geometric gauge $\varphi_{\mathrm{geo}} = \theta + 2\pi\,\mathrm{round}(s\, r/p - \theta/2\pi)$ — the winding coordinate of Eq. (1) rounded to its nearest integer turn and mapped back to an angle — anchors only strokes whose geometric coordinate agrees with their measured arc length; the reduced system solves anchors-only with a small Tikhonov term on near-null directions. Five reweighting rounds (plus two at tightened $\sigma/3$) downweight outlier rows, and pool-adjacent-violators projection restores per-stroke monotonicity after each round, which is what makes the post-cut, fragmented system well-behaved where the earlier active-set QP was not.

C.8 Retired: Discrete Winding Lift by Tabu Search

Structures. A *chart* is one connected component of the retained mesh; since charts never share vertices across cubes, a chart is one

cube's sheet. Two charts are *authentic neighbours* when at least 3 vertex pairs lie within 6 (twice the maximum bridge radius, deliberately below the 7-voxel clearance so a "neighbour" can never be the next wrap in) with contact normals agreeing to $|\cos| \geq 0.7$; a *welded group* is a connected component of that graph, i.e. one physically welded fragment. Before the search, faces whose UV/XYZ edge stretch exceeds 8× are killed, and a per-chart wind reconciliation integrates weld gauge evidence over a maximum-confidence spanning forest, using only weld edges whose turn residual is below 0.25 and capping corrections at ±8 turns.

Energy. With $k_c \in [-K, K]$ the integer wind depth of chart c, the canonical discrete energy is

$$E(k) = \lambda_o N_{\text{ov}}(k) + \lambda_n \sum_{(a,b)\in\mathcal{E}} w_{ab}\,|(k_a - k_b) - t_{ab}| + \lambda_r \sum_c H(k_c - \widehat{k}_c),$$

where N_{ov} counts exact overlapping UV triangle pairs between charts that are not authentic neighbours, $\mathcal{E}$ holds the trusted weld and lateral relations with integer targets t_{ab}, and H is a Huber penalty [Huber 1964] between the winding implied by a chart's u and the winding implied by its radius. Defaults are $\lambda_o = 1$ per exact overlapping face pair, $\lambda_n = 4$ per contact-weighted torn turn, and $\lambda_r = 2$ per turn of Huber radial deviation. Weld edges weight w_{ab} by physical contact count; lateral same-wind edges (same radius within 0.6 pitch, adjacent cubes, side by side within $\Delta u \leq 200$) carry 8 contact-equivalents and target $t_{ab} = 0$; edges inside a happy family are boosted 8×. A steering-only variant downweights edges to a currently-colliding neighbour by 0.25 when *ranking* candidate moves, but the canonical (unsteered) energy owns acceptance, best-tracking, aspiration, and stopping. The one-wind ladder — the signed Δu of moving one wind outward — is the median over radially adjacent same-u chart pairs; if too few pairs support it, the solver falls back to the calibrated spiral map. Exact overlap counting memoizes face-pair tests behind chart- and face-level AABB/sweep rejection, with cube provenance excluding pairs emitted by one cube.

Search. Depths span $k \in [-4, 4]$. Each iteration evaluates single-chart moves and the compound moves (whole welded group, lateral neighbourhood, graded rewind with bins from integrated radial climb, and compound relative-gauge regions), all in fixed-point 1/1024 energy units so no platform's floating-point summation can reorder decisions. The best non-tabu move is applied — or a tabu move, if it would beat the best energy ever seen (aspiration) — and each departed depth is banned for 40 iterations. Ties break by fixed chart order. The search stops after 300 iterations without improving the canonical best, with a hard cap of 2,000. Every accepted state is archived; a leaderboard of candidates, always including the input assignment as state 0, is re-scored by exact SAT triangle intersection, and the winner is chosen on the exact count with the summed $|k|$ as tie-break. Physical weld displacement ($|\Delta u|$ at the contact, with counts of jumps above 25 and 100) is recorded per candidate, because an integer-satisfied target can still hide a large metric discontinuity.

Relayout. With the winning depths frozen, a continuous per-chart u re-registration restores metric seam agreement: intact weld edges become closure springs, same-depth laterals keep-together springs, and radially adjacent pairs enforce at least the measured per-wind separation so the continuous stage cannot re-collapse what the search separated. Charts are weighted by face count (clipped at 0.25× the mean), deviation across any intact topology edge is budgeted at 25, and the re-layout is accepted only if an independent exact-overlap recount does not worsen; otherwise it is reverted and the pure tabu placement stands.

C.9 Retired: Quilting and Snap Terms

In E_{snap}, candidates for a dark vertex are sampled at 33 signed depths over ±8 along the across-sheet radial direction, marching in 0.25 steps and stopping at occupancy 0.9 against foreign geometry. The data term weights are 30 on the quilting mismatch $|v_{\text{RAW}} - \bar{v}|$ and 200 on the normalized displacement; the truncated-linear smoothness uses $\lambda = 2000$ with $\tau = 4$, optimized by alpha expansion. The dark-region detector is a binary graph cut with intensity threshold 64, contrast-sensitive weight $\lambda_{\text{cut}} = 20$, and $\sigma = 30$, tuned inclusive (false positives are recoverable; a missed dark island is not). The screened-Laplacian move pins clean vertices at weight 10^4, pulls island vertices toward their targets at $\alpha/(1-\alpha)$ per adjacent edge with $\alpha = 0.85$, caps displacement at 8, and reverts any vertex whose solved position violates occupancy.

C.10 Retired: Collision-Safe Ribbon Fit

Rows are sliced every 4 in v and sampled every 2 in u; duplicates merge when samples coincide within the local tolerance *and* tangents agree to $\cos \geq 0.5$, and conflicting-tangent coincidences are recorded, not averaged. Each row is fitted on a 16-voxel control spacing minimizing $\lambda_p \sum\|P(u_i) - o_i\|^2 + \lambda_s \sum\|P''\|^2 + \lambda_t \sum\|P' - \hat{t}\|^2$ with $(\lambda_p, \lambda_s, \lambda_t) = (1, 8, 4)$, where the tangent term integrates unit metric progress in u. Unsupported gaps are bridged only up to 512 and only if the chord passes the $1.25\,\Delta u + 4$ stretch test. Chart U shifts are then smoothed along v (weight 16, six bidirectional sweeps), and the exact collision test repeatedly applies 0.35 of each computed escape for up to eight rounds, stopping when the ledger stops improving.

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